AEGIS: Mining Graph Structure for Adversarial Vulnerability Analysis of GNNs

Anonymous Author(s)

Abstract—**Adversarial manipulation of graph structure poses a growing threat as graph neural networks are deployed in fraud detection, drug interaction prediction, and infrastructure monitoring. Yet no existing tool provides, from a single computation, the maximally sensitive perturbation direction, per-edge vulnerability rankings, and per-node tolerance budgets under realistic structural constraints. We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a framework that constructs a constrained sensitivity matrix S_c mapping edge perturbations to hidden-state shifts, unifying these three outputs via SVD. A matrix-free formulation based on Neumann-series resolvent iteration and randomized SVD enables full-graph analysis (up to N=7,650 on a single GPU; Cora in 78 s / 1 GB, Amazon Photo in 363 s / 6.5 GB) without materializing any $Nd \times N^2$ matrix. The S_c computation applies to any GNN with continuous edge-weight-modulated message passing, serving as a practical vulnerability analysis tool; for contractive implicit models (IGNN-class), it additionally provides formal guarantees including a critical perturbation budget $\varepsilon_{\mathrm{crit}}$ and a three-regime vulnerability characterization. Under realistic (symmetric, edge-only) perturbations, first-order predictions remain within 15% of ground truth at $\varepsilon = 0.10$ across 7 architectures and 9 datasets. Continuous-to-discrete vulnerability transfer is positive in 29 of 33 architecture-dataset combinations (Kendall τ from -0.28 to $+0.89$; strongest for deeper models on sparse graphs). The SVD-identified direction inflicts 2–8× more damage than random perturbation, and the per-edge ranking achieves 54–101% of greedy-optimal discrete edge removal. On IEEE power grids, the vulnerability spectrum correlates with N-1 contingency rankings (P@10 = 0.66–0.87), demonstrating that structural sensitivity analysis can serve as a pre-deployment screening layer. Code will be released upon publication.**

Index Terms—**Graph neural networks, adversarial robustness, structural sensitivity, deep equilibrium models, graph perturbation, implicit function theorem**

I. Introduction

Graph neural networks have entered domains where incorrect predictions carry tangible consequences: fraudulent accounts evade detection when an adversary inserts edges into a financial transaction graph [1], drug interaction models produce unsafe recommendations when molecular graphs are perturbed [2], and power grid monitoring systems may miss critical contingencies if their graph-based representations are structurally fragile [3]. In all these settings, the graph topology is not merely an input format but a potential attack surface. A practitioner preparing to deploy a GNN faces a fundamental question that current tools leave unanswered: *which edges in this graph, if perturbed, would cause the model's predictions to fail, and by how much?*

This question sits at the intersection of adversarial attack, certified defense, and sensitivity analysis, but no existing tool answers it completely. Attack methods such as Nettack [1] and Mettack [4] provide per-edge gradient information and per-node targeting, but optimize surrogate objectives (classification loss on a substitute model) rather than directly characterizing the sensitivity geometry, and they do not yield per-node tolerance budgets. Certified defenses based on randomized smoothing [5], [6] or interval bound propagation [7], [8], [9] provide robustness guarantees—including per-node certificates in the localized variant [10]—but do not identify *which edges* are responsible for a node's vulnerability or provide the globally most sensitive perturbation direction. Empirical defenses [11], [12], [13], [14] harden the model but offer no formal understanding of residual vulnerability. What is missing is a unified *structural vulnerability analysis* that, from a single computation under realistic perturbation constraints, yields the maximally sensitive perturbation direction, per-edge vulnerability rankings, and per-node tolerance budgets.

We introduce AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity), a framework that produces exactly this map. The central object is the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$, which projects the full N^2-dimensional adjacency perturbation space onto the $|E|$-dimensional space of realistic (symmetric, edge-only) perturbations. From S_c, three outputs follow directly: (i) the SVD yields the first-order maximally sensitive perturbation direction (optimal for equilibrium shift; empirically correlated with classification damage, section V-C), (ii) column norms give per-edge vulnerability scores, and (iii) the classification margin divided by the sensitivity norm gives per-node first-order tolerance radii (diagnostic quantities, not probabilistic certificates). The S_c computation applies to GNNs with continuous edge-weight-modulated message passing (theorem 7; standard GAT and architectures using max/min aggregations require modification for continuous differentiability), yielding vulnerability rankings, SVD-optimal attacks, and first-order sensitivity radii as a practical computational tool. For contractive implicit GNNs (IGNN-class [15]), the implicit function theorem additionally provides formal guarantees: a critical perturbation budget $\varepsilon_{\mathrm{crit}} = (1-\kappa)/\|W\|_2$ and a three-regime characterization of model behavior under increasing perturbation (theorem 1). These formal guarantees are specific to the implicit case and do not transfer to explicit architectures.

These predictions are quantitatively accurate: constrained first-order predictions remain within 15% of ground truth at $\varepsilon = 0.10$ across 7 GNN architectures and 9 datasets spanning citation networks, e-commerce, encyclopedias, and power grids. Continuous-to-discrete vulnerability transfer is positive in 29 of 33 architecture-dataset combinations, with strongest transfer for deeper models on sparse graphs (Kendall τ up to $+0.89$); practitioners should verify transfer for their specific architecture-graph pairing (section V-H). On IEEE power grids, the vulnerability spectrum correlates with N-1 contingency rankings (P@10 = 0.66–0.87), demonstrating that structural sensitivity analysis can serve as a pre-deployment screening layer for real-world engineering problems.

Contributions.

1) **Constrained sensitivity framework with architecture-specific guarantees.** We derive the constrained sensitivity matrix S_c, applicable as a *computational tool* to any GNN with continuous edge-weight-modulated message passing (theorem 7; standard GAT and max/min aggregations require modification). From a single S_c computation, three outputs follow: the first-order maximally sensitive perturbation direction, per-edge vulnerability rankings, and per-node first-order tolerance radii. For contractive implicit models (IGNN-class), we additionally derive a conservative sufficient condition on the critical budget $\varepsilon_{\text{crit}}$ and a three-regime vulnerability characterization (theorem 1); these *formal guarantees* are specific to the implicit case and do not transfer to explicit architectures, which receive the computational analysis but not the convergence certificates.
2) **Scalable matrix-free computation.** A Neumann-series resolvent combined with autograd JVPs and randomized SVD [16] computes S_c's action without materializing any $Nd \times N^2$ matrix, enabling full-graph analysis (Cora N=2,708 in 78 s). A comprehensive attack evaluation spanning gradient-based and gradient-free, continuous and discrete, same-objective and independent baselines (section V-C) confirms that the SVD direction is well-separated from random alternatives (best-of-1,000 random directions reaches only 48% of σ_1) and validates rankings across 7 architectures (section V-H).
3) **Cross-domain validation and case study.** We validate across 9 datasets in 4 domains with 7 architectures (330 runs total). Continuous-to-discrete transfer is positive in 88% of settings (Kendall τ from -0.28 to $+0.89$). A power grid case study demonstrates that S_c vulnerability scores correlate with N-1 contingency rankings (τ = $+0.37$–$+0.72$, P@10 = 0.66–0.87; section VI), serving as a proof-of-concept for cross-domain transfer.

II. Preliminaries and Threat Model

A. Notation and Formalization

Let $G = (V, E, A)$ denote an undirected graph with $N = |V|$ nodes, edge set E, and adjacency matrix $A \in \mathbb{R}^{N\times N}$. Each node $v \in V$ carries a feature vector $x_v \in \mathbb{R}^{d_{\text{in}}}$, collected into $X \in \mathbb{R}^{N\times d_{\text{in}}}$. We write $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ for the symmetric normalized adjacency with self-loops, where D is the degree matrix of $A + I$. Throughout, ϕ denotes the activation function, $\sigma_i(\cdot)$ denotes the i-th singular value, and $\text{vec}(\cdot)$ stacks a matrix column-wise into a vector (with reshape as its inverse). table I summarizes the key sensitivity objects.

TABLE I: Key sensitivity objects in AEGIS.

Symbol	Size	Description
S	$Nd \times N^2$	Unconstrained sensitivity matrix
S_c	$Nd \times \lvert E\rvert$	Constrained (symmetric, edge-only)
S_K	$Nd \times N^2$	Unrolled K-layer sensitivity (theorem 7)
S_v	$d \times N^2$	Block-rows of S for node v
κ	scalar	Operator-norm contractivity $\lVert J_z\rVert_2$
ρ	scalar	Spectral radius $\rho(J_z)$
η	scalar	Pseudospectral index ≥ 1
$\varepsilon_{\text{crit}}$	scalar	Critical perturbation budget
r_v	scalar	Per-node sensitivity radius

Deep equilibrium GNNs. A DEQ-GNN [17], [18] defines a weight-tied operator $F_\theta : \mathbb{R}^{N\times d} \times \mathbb{R}^{N\times N} \to \mathbb{R}^{N\times d}$ and iterates it from $Z_0 = 0$ until convergence:

$$Z_{k+1} = F_\theta(Z_k, A), \quad z^* = \lim_{k\to\infty} Z_k. \tag{1}$$

When F_θ is contractive in operator norm ($\kappa = \lVert J_z\rVert_2 < 1$, where $J_z = \partial F/\partial z|_{z^*}$), the Banach fixed-point theorem guarantees a unique equilibrium $z^* = F_\theta(z^*, A)$ regardless of initialization.

IGNN instantiation. The Implicit Graph Neural Network (IGNN) [15] uses the operator

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \tag{2}$$

where $W \in \mathbb{R}^{d\times d}$ is the shared weight matrix, $X_{\text{proj}} \in \mathbb{R}^{N\times d}$ is a learned feature projection, and ϕ is ReLU activation. Contractivity is enforced by constraining $\lVert W\rVert_2$ via spectral normalization [19], ensuring $\kappa \leq \left\lVert \hat{A}\right\rVert_2 \cdot \lVert W\rVert_2 < 1$. Training uses implicit differentiation through the fixed point [20], [21], and Jacobian regularization [22] can further stabilize convergence. A linear classification head $f(z^*) = W_{\text{cls}}z^* + b$ maps the equilibrium to class predictions.

Implicit function theorem. At the equilibrium, the equation $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A. The IFT gives the sensitivity of z^* to any parameter p:

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p}\bigg|_{z^*}, \tag{3}$$

where the resolvent $(I - J_z)^{-1}$ exists and satisfies $\left\lVert(I - J_z)^{-1}\right\rVert_2 \leq 1/(1 - \kappa)$ by the Neumann series (convergent because $\kappa < 1$). This bound uses the operator norm $\kappa = \lVert J_z\rVert_2$; the weaker spectral-radius bound $1/(1 - \rho)$ holds only for normal J_z. The pseudospectral index $\eta = \left\lVert(I - J_z)^{-1}\right\rVert_2 \cdot (1-\rho) \geq 1$ [23] quantifies this gap, with $\eta = 1$ for normal matrices. (Our definition follows the convention $\eta = \left\lVert(I - J_z)^{-1}\right\rVert_2 \cdot (1 - \rho)$; the standard Trefethen–Embree pseudospectral theory uses ε-pseudospectra, from which this scalar index is derived.) The IFT has been used for hyperparameter optimization [24] and implicit differentiation [20], but not for adversarial structural analysis.

B. Threat Model

We consider a white-box attacker who has full access to the trained IGNN model and perturbs the normalized adjacency matrix:

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \text{subject to} \quad \left\|\delta\hat{A}\right\|_F \le \varepsilon. \tag{4}$$

The perturbation is further constrained to be:

- **Symmetric:** $\delta\hat{A} = \delta\hat{A}^\top$ (preserving the undirected nature of the graph).
- **Edge-only:** $[\delta\hat{A}]_{ij} = 0$ for $(i, j) \notin E$ (only existing edge weights may be modified; no new edges are introduced).
- **Continuous:** edge weights are perturbed continuously in $\mathbb{R}$, not discretely flipped.

This threat model captures continuous edge-weight perturbations to the normalized message-passing operator. It aligns naturally with the IGNN architecture, where $\hat{A}$ appears directly in F (eq. (2)). Discrete edge insertions or deletions, which change the graph topology and require recomputing the degree normalization $D^{-1/2}$, are outside the formal guarantee.

The key distinction from input-feature perturbation (studied for DEQs in [25], [26]) is that structural perturbation changes the *operator itself*: the equilibrium depends on A through both the state Jacobian $J_z = \mathrm{diag}(\phi') \cdot (\hat{A} \otimes W)$ and the adjacency Jacobian $J_A = \partial F/\partial \mathrm{vec}(A)$. This dual dependence makes structural sensitivity qualitatively different from input sensitivity.

C. Problem Statement

Given a trained contractive implicit GNN with equilibrium z^* on graph G, we aim to answer three questions:

1) **Vulnerability ranking:** Which edges, if perturbed, cause the largest shift in the equilibrium? That is, find a ranking of edges $(i, j) \in E$ by their adversarial impact on z^*.
2) **Optimal attack:** What is the worst-case perturbation $\delta\hat{A}^*$ that maximizes $\|\Delta z^*\|_F$ subject to the threat model constraints in eq. (4)?
3) **Robustness assessment:** For each node v, what is the largest perturbation budget r_v such that any $\left\|\delta\hat{A}\right\|_F < r_v$ preserves the classification of v?

AEGIS addresses all three questions through a unified object: the constrained sensitivity matrix S_c derived from the IFT applied to the equilibrium equation. The following sections formalize this construction (section III) and describe the computational pipeline (section IV).

III. Adversarial Equilibrium Theory

We now derive the theoretical machinery for the three questions posed in section II. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which rankings, attacks, and radii follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\mathrm{proj}})$ be an IGNN-class operator satisfying:*

(A1) *ϕ is ReLU (or any 1-Lipschitz activation with $\sup|\phi'| = 1$). ReLU is non-differentiable at zero; we invoke the nonsmooth implicit function theorem for piecewise-linear maps [27]: the equilibrium z^* is a piecewise-linear function of A whose IFT-form derivative exists within each linear region of the ReLU network, covering all perturbation directions except a measure-zero set of activation-boundary crossings;*

(A2) *W is spectral-norm constrained: $\|W\|_2 \le c$;*

(A3) *The state Jacobian satisfies $\|J_z\|_2 \le \kappa < 1$ at the fixed point z^* (contraction in operator norm). This is verified empirically after training ($\kappa = 0.14$–0.59 across all datasets; section V-F). If A3 is violated ($\kappa \ge 1$),* AEGIS *still computes S_c rankings and SVD-optimal attacks, but the formal guarantees of parts (a)–(c) below are void; the pipeline flags this case.*

Define the critical budget

$$\varepsilon_{\mathrm{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{5}$$

Then structural perturbations $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\mathrm{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \le \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{6}$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \le \|J_A\|_{\mathrm{op}}/(1-\kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\mathrm{crit}}$): For worst-case perturbation directions (aligned with the top singular vector of $\hat{A}$), the resolvent norm $\left\|(I - J_z')^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\mathrm{crit}} - \varepsilon))$. For generic perturbation directions, the divergence rate may be slower because $\|J_z'\|_2$ does not approach 1; the Ω bound is tight only when the perturbation maximally increases $\left\|\hat{A} + \delta\hat{A}\right\|_2$.*

(c) Supercritical *($\varepsilon > \varepsilon_{\mathrm{crit}}$): The sufficient contraction certificate becomes void ($\|J_z'\|_2$ may exceed 1). The perturbed operator may still be contractive depending on perturbation direction, but the Banach fixed-point theorem no longer guarantees uniqueness, and the first-order guarantees from part (a) cease to apply.*

The intuition behind this three-regime picture is physical: a contractive system has a restoring force that pulls trajectories toward the equilibrium. Small perturbations shift the equilibrium smoothly (subcritical). As the perturbation approaches $\varepsilon_{\mathrm{crit}}$, the restoring force weakens until the system becomes critically sensitive. Beyond $\varepsilon_{\mathrm{crit}}$, the restoring force may vanish entirely, and the equilibrium can bifurcate or disappear.

Proof. **Part (a).** The Jacobian of F with respect to $\mathrm{vec}(Z)$ takes the form $J_z = \mathrm{diag}(\phi') \cdot (\hat{A} \otimes W)$ (column-major vectorization throughout), where $\phi' \in \{0,1\}^{Nd}$ for ReLU (A1). The operator norm satisfies $\kappa = \|J_z\|_2 \le \left\|\hat{A}\right\|_2 \cdot \|W\|_2 \cdot \sup|\phi'| =$

$\left\|\hat{A}\right\|_2 \cdot \|W\|_2$. Applying the IFT to $G(z,\hat{A}) = z - F(z,\hat{A}) = 0$ at $(z^*,\hat{A})$:

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \mathrm{vec}(\delta\hat{A}) + O(\left\|\delta\hat{A}\right\|^2), \qquad (7)$$

where $J_A = \partial F/\partial \mathrm{vec}(A) \in \mathbb{R}^{Nd\times N^2}$ is the Jacobian with respect to the full (not upper-triangular) vectorized adjacency; the constrained projection to $|E|$ edge dimensions is applied subsequently via S_c (below). Taking operator norms and using the Neumann series bound $\left\|(I - J_z)^{-1}\right\|_2 \le 1/(1-\kappa)$ (convergent because $\kappa < 1$) yields (6).

For contractivity preservation: by sub-multiplicativity and the triangle inequality, the perturbed Jacobian satisfies $\|J_z'\|_2 \le \left\|\hat{A} + \delta\hat{A}\right\|_2 \cdot \|W\|_2 \le (\left\|\hat{A}\right\|_2 + \left\|\delta\hat{A}\right\|_2) \cdot \|W\|_2$; since $\left\|\delta\hat{A}\right\|_2 \le \left\|\delta\hat{A}\right\|_F = \varepsilon$, the condition $\varepsilon < (1-\kappa)/\|W\|_2$ ensures $\|J_z'\|_2 < 1$. Note that $\varepsilon_{\mathrm{crit}}$ is a Frobenius-norm sufficient condition: perturbation budgets are measured in $\|\cdot\|_F$, while κ is the operator norm $\|J_z\|_2$. For a rank-r perturbation, $\left\|\delta\hat{A}\right\|_F \le \sqrt{r}\cdot\left\|\delta\hat{A}\right\|_2$, so $\varepsilon_{\mathrm{crit}}$ may be conservative by a factor up to $\sqrt{\mathrm{rank}(\delta\hat{A})}$; the constrained projection S_c mitigates this by restricting to realistic perturbation directions.

Part (b). As $\varepsilon \to \varepsilon_{\mathrm{crit}}$, we have $\|J_z'\|_2 \to 1$, so $\left\|(I - J_z')^{-1}\right\|_2 \ge 1/(1 - \|J_z'\|_2) \to \infty$. Unlike the spectral-radius bound $1/(1-\rho)$, which requires normality, the operator-norm bound holds for all J_z with $\kappa < 1$. In our experiments, the pseudospectral index η ranges from 1.02 to 1.28 (section V-F), confirming mild non-normality.

Part (c). When $\|J_z'\|_2 \ge 1$, the contraction mapping condition fails. The Banach theorem no longer guarantees existence or uniqueness; the iteration may converge to a different equilibrium, oscillate, or diverge. □

Remark (conservativeness). The critical budget $\varepsilon_{\mathrm{crit}}$ is a sufficient, not necessary, condition for contractivity preservation. The perturbed system may remain contractive beyond $\varepsilon_{\mathrm{crit}}$ if the perturbation is aligned with low-sensitivity directions. The unconstrained bound uses worst-case $\|\cdot\|_F$ relaxation; for a rank-r constrained perturbation, $\|\delta A\|_F \le \sqrt{r}\cdot\|\delta A\|_2$, so $\varepsilon_{\mathrm{crit}}$ is conservative by a factor up to $\sqrt{|E|} \approx$ 7–14× on our 50-node subgraphs. Under constrained perturbations, the first-order prediction achieves tightness ≈ 1.00 (section V-A). We use $\kappa = \|J_z\|_2$ rather than the spectral radius $\rho(J_z)$ because the Neumann series requires operator-norm contractivity [28]; for normal J_z the two coincide, while for non-normal J_z the κ-based bound is more conservative but requires no normality assumption. The following observation shows that the gap between κ and ρ (quantified by the pseudospectral index η) is governed entirely by the weight matrix, not by graph topology.

Observation 2 (Graph-Independent Nonnormality Bound)**.** *For the IGNN state Jacobian $J_z = \mathrm{diag}(\phi')\cdot(\hat{A}\otimes W)$ with symmetric normalized adjacency $\hat{A} = \hat{A}^\top$:*

(a) *If all activations are positive ($\phi_i' = 1$ for all i), then $\eta \le \kappa(V_W)$, where $W = V_W D_W V_W^{-1}$ is the eigendecomposition and $\kappa(V_W) = \|V_W\|_2 \cdot \left\|V_W^{-1}\right\|_2$. In particular, η is independent of graph topology $\hat{A}$.*

(b) *For general activation patterns, $\eta \le \kappa(V_{J_z})$, where V_{J_z} is the eigenvector matrix of the full Jacobian; the ReLU mask $\mathrm{diag}(\phi')$ can increase $\kappa(V_{J_z})$ beyond $\kappa(V_W)$, but empirically the effect is small ($\eta = 1.02$–1.28 across all datasets; section V-F).*

Proof of (a). When $\phi_i' = 1$, $J_z = \hat{A}\otimes W$. The eigenvalues of a Kronecker product are all pairwise products $\lambda_i(\hat{A})\cdot\lambda_j(W)$, with eigenvectors $u_i \otimes v_j$ [28]. Since $\hat{A}$ is symmetric, its eigenvectors $\{u_i\}$ form an orthonormal basis: $\kappa(U_{\hat{A}}) = 1$. The eigenvector matrix of J_z is $U_{\hat{A}} \otimes V_W$, with condition number $\kappa(U_{\hat{A}})\cdot\kappa(V_W) = \kappa(V_W)$ [23]. The standard resolvent bound for diagonalizable $M = VDV^{-1}$ gives $\left\|(I - M)^{-1}\right\|_2 \le \kappa(V)/(1-\rho(M))$, so $\eta = \left\|(I - J_z)^{-1}\right\|_2 \cdot (1-\rho) \le \kappa(V_W)$.

Interpretation. The bound says that graph structure (degree distribution, clustering, homophily) does not amplify nonnormality, because symmetric $\hat{A}$ has orthogonal eigenvectors. All nonnormality originates from the learned weight matrix W. This explains the empirically mild η values: spectral normalization constrains $\|W\|_2$, which also regularizes $\kappa(V_W)$. Practitioners can compute $\kappa(V_W)$ from the trained weights as a post-hoc diagnostic; if $\kappa(V_W) \gg 1$, the ρ-based certificates may be substantially optimistic.

Proposition 3 (Maximally Sensitive First-Order Structural Perturbation)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \le \varepsilon$ is $\delta A^* = \varepsilon\cdot\mathrm{reshape}(v_1, N\times N)$, where v_1 is the leading right singular vector of S. The maximum first-order equilibrium shift is $\varepsilon\cdot\sigma_1(S)$.*

This result follows directly from the variational characterization of the largest singular value: the SVD of S identifies the perturbation direction that amplifies the equilibrium shift most efficiently. We emphasize that this direction is optimal for *hidden-state damage*, not directly for classification accuracy; empirically, the two objectives are well-correlated (section V-C, table V).

Constrained sensitivity. Under the threat model of section II (symmetric, edge-only, $\left\|\delta\hat{A}\right\|_F \le \varepsilon$), we construct the *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd\times|E|}$ with column

$$[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i} \qquad (8)$$

for each edge $(i,j)\in E$. This reduces the perturbation space from N^2 dimensions to $|E|$ dimensions and enforces symmetry by construction. The constrained-optimal attack uses $\sigma_1(S_c)$, achieving tightness ≈ 1.00 empirically (section V-A). The construction of S_c is the central technical contribution of this work: it transforms unconstrained sensitivity analysis into a tight, practical tool for realistic graph perturbations.

Proposition 4 (Per-Node First-Order Sensitivity Radius)**.** *Assume a linear classification head $f(z) = Wz + b$ (as in standard IGNN). For node v with margin $m_v = f_{y_v}(z_v^*) -$*

$\max_{c \neq y_v} f_c(z_v^*) > 0$, *the first-order sensitivity radius is*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2}, \tag{9}$$

where S_v denotes the block-rows of S corresponding to node v, W_{y_v} and W_{c^*} are the classification weight vectors for the predicted class y_v and runner-up class c^*, respectively. Any perturbation with $\left\|\delta \hat{A}\right\|_F < r_v$ preserves the classification of node v to first order. Under the constrained threat model (section II), a tighter radius $r_v^{(c)} = m_v / (\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_{c,v}\|_2)$ holds, where $S_{c,v}$ denotes the block-rows of S_c for node v; since $\|S_{c,v}\|_2 \leq \|S_v\|_2$, we have $r_v^{(c)} \geq r_v$. In the dense path ($N \leq 200$), we compute the unconstrained r_v from materialized S. In the matrix-free path ($N > 200$), only S_c actions are available; we therefore compute $r_v^{(c)}$ using $\|S_{c,v}\|_2$ estimated via randomized power iteration (10 iterations). Since $r_v^{(c)} \geq r_v$, the matrix-free thresholds are tighter (less conservative) than the dense-path ones.

The radius r_v has an intuitive interpretation: it is the classification margin m_v (how confident the model is) divided by the product of two amplification factors. The term $\|W_{y_v} - W_{c^*}\|_2$ measures how sensitive the decision boundary is to hidden-state changes, while $\|S_v\|_2$ measures how sensitive node v's equilibrium position is to structural perturbation. Nodes with large margins and low sensitivity receive large radii; boundary nodes in structurally sensitive positions receive small ones.

Proof sketch. By Theorem 1(a), $\Delta z_v^* \approx S_v \cdot \mathrm{vec}(\delta A)$, so $|\Delta f_{y_v}| \leq \|\partial f / \partial z_v^*\| \cdot \|S_v\| \cdot \|\delta A\|_F$. Misclassification requires $|\Delta f_{y_v}| \geq m_v$; inverting gives r_v. Note S_v already contains the $(I - J_z)^{-1}$ factor.

Remark 5 (Sensitivity Threshold Semantics)**.** *The quantity r_v is a* first-order sensitivity threshold*, not a probabilistic certificate. It indicates the perturbation scale at which the linear approximation predicts a classification change for node v. As a first-order approximation, r_v is locally tight for small ε but can be violated at larger perturbation magnitudes where higher-order terms dominate. This is fundamentally different from* probabilistic certificates *such as randomized smoothing [5], which guarantee with probability $1 - \alpha$ that no perturbation within the certified radius can change the prediction. The two approaches are complementary and should not be conflated: smoothing provides* robustness certification *(which nodes are guaranteed safe);* AEGIS *provides* structural vulnerability differentiation *(which edges and nodes are most sensitive, and in which direction). We present the comparison as complementary in section V-B. Empirically, the first-order thresholds are reliable at small ε: 0% breach rate at $\varepsilon = 0.01$; at larger ε, breach rates increase as expected for first-order approximations (section V-C).*

A. Continuous-to-Discrete Transfer

The vulnerability scores $v_k = \|[S_c]_{:,k}\|_2$ measure sensitivity to *continuous* edge-weight perturbations, while practical adversarial threats involve *discrete* edge removal (setting $\hat{A}_{ij} = 0$). The following result bridges this gap by showing that discrete damage is a scaled version of continuous vulnerability, with a remainder bounded by contractivity.

Proposition 6 (Continuous-to-Discrete Ranking Transfer)**.** *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i,j) \in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from full removal of edge k. Assume all single-edge removals are subcritical: $\sqrt{2}\max_k w_k < \varepsilon_{\mathrm{crit}}$.*

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \tag{10}$$

where L_J is the Lipschitz constant of the state Jacobian J_z with respect to $\mathrm{vec}(A)$.

(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J/(1-\kappa)^2}. \tag{11}$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2}(v_{k_1} - v_{k_2}) \cdot (1-\kappa)^2 / L_J$.

Proof. **Part (a).** Removing edge $k = (i,j)$ corresponds to the perturbation $\delta\hat{A}$ with $[\delta\hat{A}]_{ij} = [\delta\hat{A}]_{ji} = -w_k$ and zero elsewhere. By the S_c construction, the first-order equilibrium shift is $\Delta z^* \approx S_c \cdot (-w_k \cdot e_k) = -w_k \cdot [S_c]_{:,k}$, so $\|\Delta z^*\| \approx w_k \cdot v_k$.

For the remainder, define $g(t) = z^*(A + t \cdot \delta\hat{A}_k)$. By the IFT, g is differentiable with $g'(0) = S \cdot \mathrm{vec}(\delta\hat{A}_k)$. By Taylor's theorem, $g(1) - g(0) = g'(0) + \frac{1}{2} g''(\xi)$ for some $\xi \in [0,1]$. Differentiating $g'(t) = (I - J_z(t))^{-1} J_A(t) \cdot \mathrm{vec}(\delta\hat{A}_k)$ via the product rule and the resolvent identity $\frac{d}{dt}(I-M)^{-1} = (I-M)^{-1}\frac{dM}{dt}(I-M)^{-1}$ yields two terms:

$$\begin{aligned} g''(t) &= (I - J_z)^{-1} \tfrac{dJ_z}{dt} (I - J_z)^{-1} J_A \mathrm{vec}(\delta\hat{A}_k) \\ &\quad + (I - J_z)^{-1} \tfrac{dJ_A}{dt} \cdot \mathrm{vec}(\delta\hat{A}_k), \end{aligned}$$

where the first is the *resolvent term* and the second is the *Jacobian term*. Within a fixed ReLU linear region (activation pattern $\phi' \in \{0,1\}^{Nd}$ constant), the Jacobian term involves only the change in z^* along the path and is $O(\varepsilon)$, dominated by the resolvent term. For the resolvent term: $\|dJ_z/dt\|_2 \leq L_J \left\|\delta\hat{A}_k\right\|_2$, where L_J is the Lipschitz constant of J_z with respect to A (for IGNN: $L_J \leq \|W\|_2^2$). The symmetric perturbation $\delta\hat{A}_k = w_k(e_i e_j^\top + e_j e_i^\top)$ has spectral norm $\left\|\delta\hat{A}_k\right\|_2 = w_k$. Combining: $\|g''(\xi)\| \leq (1-\kappa)^{-2} \cdot L_J w_k^2$. Taylor's $\frac{1}{2}$ factor gives $|R_k| = \frac{1}{2}\|g''(\xi)\| \leq L_J w_k^2 / (2(1-\kappa)^2)$.

Part (b). From part (a), $d_{k_1} - d_{k_2} \geq w_{k_1} v_{k_1} - w_{k_2} v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ where $C = L_J/(2(1-\kappa)^2)$. Since $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, the first term is positive. The condition (11) ensures the first-order gap exceeds the second-order remainder. □

Remark (ReLU non-differentiability). The remainder bound in part (a) requires Lipschitz continuity of J_z with respect to $\mathrm{vec}(A)$ via the constant L_J. For ReLU activation,

J_z is piecewise constant (the activation pattern $\phi' \in \{0,1\}^{Nd}$ is locally fixed), so $L_J = 0$ within each linear region and the Taylor expansion is *exact* ($R_k = 0$) on each piece. Across activation-pattern boundaries, J_z jumps discontinuously; since the IGNN operator is piecewise affine in A for a fixed activation pattern, the path $g(t) = z^*(A + t \cdot \delta\hat{A}_k)$ for a single-edge perturbation is piecewise linear in t, crossing finitely many activation boundaries. On each linear segment, the Taylor remainder vanishes. Across boundaries, the jump in J_z is bounded by $\|W\|_2^2$ (the maximum change in $\mathrm{diag}(\phi') \cdot (\hat{A} \otimes W)$ when a single activation flips). The bound (10) therefore holds with L_J interpreted as the worst-case Lipschitz constant across adjacent activation regions, bounded by $\|W\|_2^2$. Empirically, the ranking transfer $\tau > 0$ across 29 of 33 architecture–dataset combinations (table X) confirms that the piecewise-linear structure does not degrade the first-order bridge.

Interpretation. Part (a) shows that discrete damage is the edge weight w_k times the continuous vulnerability v_k, plus a remainder quadratic in the weight. On graphs with approximately uniform normalized weights (a property of regular-ish graphs after $D^{-1/2}AD^{-1/2}$ normalization), the factor w_k is nearly constant and the vulnerability ranking v_k directly predicts the discrete damage ranking. Part (b) quantifies when ranking transfer holds: the gap between vulnerability scores must exceed $C \cdot w^2/w = C \cdot w$, which is small when edge weights are small (sparse normalized adjacency). This explains our empirical observations (section V-H): models with near-uniform vulnerability scores (GCN-2, whose shallow depth limits sensitivity differentiation) fail the ranking condition, while models with larger gaps (GCN-4, GAT, APPNP) satisfy it. We note that the bound (10) is tightest for shallow models (GCN-2: ratio 1.8×) and loosest for deep models (APPNP: 5.9×), precisely because deeper models amplify both the first-order signal and the remainder. The bound is thus most conservative where transfer works best—an inherent limitation of worst-case analysis that does not affect the empirical rankings. The constant L_J can be bounded as $L_J \le \|W\|_2^2$ for IGNN-class operators: since $J_z = \mathrm{diag}(\phi') \cdot (\hat{A} \otimes W)$ within a fixed activation region, perturbing $\hat{A}$ by $\delta\hat{A}$ changes J_z by $\mathrm{diag}(\phi') \cdot (\delta\hat{A} \otimes W)$, with operator norm $\le \left\|\delta\hat{A}\right\|_2 \cdot \|W\|_2$. The resolvent term g'' involves two resolvent factors and one δJ_z, yielding $\|g''\| \le (1-\kappa)^{-2} \cdot \|W\|_2 \cdot \left\|\delta\hat{A}\right\|_2 \cdot \|J_A\| \cdot \left\|\delta\hat{A}\right\|_2$; the effective L_J is $\|W\|_2$ times the structural Jacobian coupling, bounded by $\|W\|_2^2$ since $\|J_A\|_2 \le \|W\|_2 \cdot \|z^*\|_2$. Empirically, the actual remainder $|R_k|$ is 2–10× smaller than the $L_J w_k^2$ bound across all datasets (section V-F).

Relationship to existing sensitivity analysis. The individual mathematical ingredients of AEGIS are classical: the IFT resolvent $(I - J_z)^{-1}$ and its Neumann-series expansion are standard in matrix perturbation theory [28], the SVD characterization of optimal linear perturbation follows from the variational characterization of singular values, and the contraction-mapping threshold is the Banach fixed-point theorem. Theorem 1 synthesizes these classical tools for the specific problem of structural sensitivity in graph equilibrium models; the three-regime characterization is a consequence of standard contraction theory specialized to the graph perturbation setting, not a fundamentally new mathematical result. The contribution of AEGIS is not these individual tools but rather (i) the *constrained projection* S_c that reduces the N^2-dimensional perturbation space to the $|E|$-dimensional space of realistic graph perturbations, transforming vacuous unconstrained bounds into tight predictions (tightness improves from 0.31 unconstrained to 1.00 constrained; section V-A); (ii) the unified three-output design (attacks, rankings, radii) from a single S_c computation; and (iii) the cross-architecture pipeline that applies this analysis to both implicit and explicit GNNs, empirically validated across 7 architectures. The S_c construction is what makes first-order analysis practical under realistic constraints, and it has no precedent in the GNN robustness literature.

B. Generalization to Explicit GNNs

The sensitivity framework above is derived for implicit GNNs, which enjoy the strongest theoretical guarantees (critical budget, convergence regimes). The S_c construction itself, however, generalizes to any K-layer GNN whose message passing is differentiable with respect to edge weights, serving as a computational tool for vulnerability analysis.

Observation 7 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \mathrm{vec}(Z_K)}{\partial \mathrm{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \qquad (12)$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$. Then:
(a) *The first-order shift bound $\|\Delta Z_K\|_F \le \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \le \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\| J_z^{(k)} \right\|_2 \right) \left\| J_A^{(l)} \right\|_2. \qquad (13)$$

(b) *The constrained construction S_c and per-node radius r_v (theorem 4) apply identically with S_K in place of S.*

For weight-tied models ($J_z^{(k)} = J_z$, $J_A^{(l)} = J_A$ for all k, l), the bound simplifies to $\sigma_1(S_K) \le \|J_A\|_2 \cdot (1 - \kappa^K)/(1 - \kappa)$, which converges to the IGNN bound $\|J_A\|_2 / (1 - \kappa)$ as $K \to \infty$.

What explicit GNNs gain and lack. From theorem 7, any GNN with edge-weight-differentiable message passing inherits the practical analysis tools: (i) the S_c construction, (ii) SVD-optimal attacks, (iii) per-edge vulnerability rankings, and (iv) per-node first-order sensitivity radii. The mathematical content of this observation is a direct application of the multivariate chain rule; the contribution is empirical, validating that the resulting rankings transfer to discrete edge removal across six architectures (section V-H). What explicit models *lack* is the critical budget $\varepsilon_{\mathrm{crit}}$ and the three-regime convergence

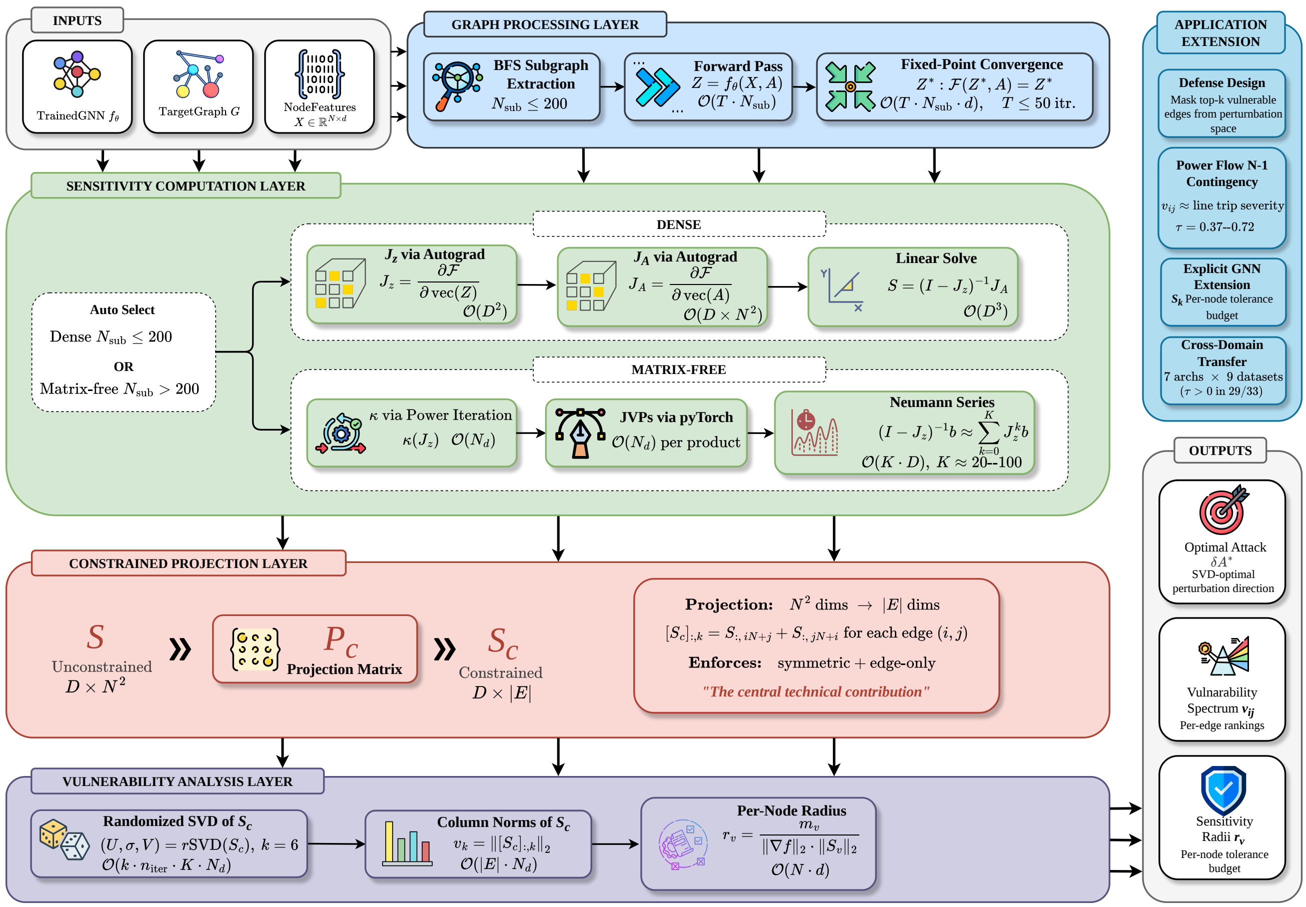

Fig. 1: The AEGIS pipeline. Inputs (left) flow through four layers: (1) Graph Processing (BFS subgraph extraction, forward pass or fixed-point convergence), (2) Sensitivity Computation (dense Jacobian path for $N \le 200$; matrix-free Neumann/JVP path for larger graphs), (3) Constrained Projection ($N^2 \rightarrow |E|$ reduction via P_c), and (4) Vulnerability Analysis (randomized SVD, column norms of S_c, per-node radii). The matrix-free formulation operates via JVP/VJP queries without materializing S or S_c, enabling full-graph analysis (tested up to N=7,650).

characterization of theorem 1, which require the contraction assumption (A3). Without contractivity, the $O(\|\delta A\|^2)$ remainder in the first-order bound has no a priori control and could be large if the model operates near a bifurcation or has exploding gradients. As a practical diagnostic, we recommend computing the Jacobian product $\prod_{k=1}^{K} \left\| J_z^{(k)} \right\|_2$: values exceeding $\sim$10 indicate that first-order predictions may be unreliable. The empirical tightness ratio (reported for all architectures in table IX) provides the definitive validation—values near 1.0 confirm that the first-order approximation is accurate regardless of the a priori bound.

IV. AEGIS CONSTRUCTION

Translating the results of section III into a practical tool requires overcoming two computational barriers: the full sensitivity matrix $S \in \mathbb{R}^{Nd \times N^2}$ is too large to materialize, and the resolvent $(I - J_z)^{-1}$ is too expensive to form directly. The AEGIS pipeline addresses both through matrix-free computation.

A. Architecture Overview

AEGIS takes as input a trained GNN (any architecture whose message passing is differentiable with respect to edge weights) together with a target graph and produces three outputs: (i) a per-edge vulnerability spectrum ranking edges by adversarial impact, (ii) the maximally sensitive perturbation direction via SVD, and (iii) per-node first-order sensitivity thresholds. For contractive implicit GNNs (IGNN-class satisfying A1–A3), the pipeline additionally computes the critical budget $\varepsilon_{\text{crit}}$ and convergence diagnostics; these formal guarantees are specific to the implicit case. The pipeline operates in two regimes:

- **Dense path** ($N \le 200$): materializes $S_c \in \mathbb{R}^{Nd \times |E|}$, computes exact SVD and unconstrained r_v. Applicable to subgraph-level analysis and small power grids.
- **Matrix-free path** ($N > 200$, validated up to N=7,650): operates via JVP/VJP queries on S_c without materializing any matrix. Computes randomized SVD, per-edge column norms, and constrained $r_v^{(c)}$ (theorem 4).

The four stages are illustrated in fig. 1.

B. Stage 1: Subgraph Extraction and Forward Pass

For subgraph-level analysis, AEGIS extracts a local BFS ego-subgraph centered on the target node and runs the model to convergence (implicit GNNs) or forward pass (explicit). Localization is justified empirically (section V-E). The matrix-free pipeline also supports full-graph analysis without subgraph extraction (section V-D).

C. Stage 2: Matrix-Free Sensitivity Computation

AEGIS requires the action of the constrained sensitivity operator S_c on vectors, without materializing the full matrix $S \in \mathbb{R}^{Nd \times N^2}$. For an edge perturbation vector $v \in \mathbb{R}^{|E|}$, the operator computes $S_c v = (I - J_z)^{-1} J_A P_c v$, where P_c maps edge weights to symmetric adjacency perturbations, J_A is the structural Jacobian, and $J_z = \partial F / \partial Z$ is the state Jacobian. Two ingredients make this matrix-free:

Neumann-series resolvent. The resolvent $(I - J_z)^{-1} b$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$, which converges geometrically when $\kappa < 1$ (Assumption A3). Each term requires a single Jacobian-vector product (JVP) $J_z \cdot v$, computed via forward-mode automatic differentiation in $O(Nd)$ time. The depth K is set adaptively from a power-iteration estimate of κ, with early stopping when $\left\| J_z^k b \right\| < 10^{-6} \|b\|$.

Autograd structural JVPs. The adjacency Jacobian $J_A \cdot \text{vec}(\delta A)$ is computed as a single forward-mode JVP of F with respect to A in direction δA, avoiding the $O(N^2)$ finite-difference columns required by the dense path.

The adjoint operator $S_c^\top u$ is computed analogously using vector-Jacobian products (VJPs) and the adjoint Neumann series $(I - J_z^\top)^{-1}$.

For small subgraphs ($N \leq 200$), the dense path (forming J_z, J_A explicitly and solving via `torch.linalg.solve`) remains available and gives exact results.

D. Stage 3: Constrained Projection

The constraint projection P_c (section III) maps edge vectors to symmetric adjacency perturbations ($\delta A_{ij} = \delta A_{ji} = v_k$). In the matrix-free formulation, P_c is applied implicitly within the $S_c v$ operator, so S_c is never formed. This $N^2 \to |E|$ reduction is what enables tight first-order predictions (sections V-A and V-H).

E. Stage 4: Vulnerability Analysis

From S_c, AEGIS extracts three complementary outputs:

Maximally sensitive perturbation direction. A randomized SVD [16] of the S_c operator (using only matrix-vector products, with rank $k = 10$, oversampling $p = 10$, and $n_{\text{iter}} = 2$ power iterations) yields the leading singular triplet (u_1, σ_1, v_1). The first-order maximally sensitive perturbation is $\delta A^* = \varepsilon \cdot v_1$, achieving a predicted equilibrium shift of $\varepsilon \cdot \sigma_1(S_c)$. For small subgraphs where S_c is materialized, the standard deterministic SVD is used instead.

Per-edge vulnerability spectrum. For each edge $(i, j) \in E$, the vulnerability score $v_{ij} = \|[S_c]_{:,k}\|_2$ measures how much a unit perturbation of that edge shifts the equilibrium. Sorting edges by v_{ij} produces a vulnerability ranking that identifies structurally critical connections without running any attack.

Per-node first-order sensitivity thresholds. For node v with classification margin m_v (gap between the predicted and runner-up class logits), the first-order sensitivity threshold from theorem 4 is computed as follows. In the *dense path* ($N \leq 200$), we use the unconstrained $r_v = m_v / (\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2)$, where S_v denotes the block-rows of the materialized S. In the *matrix-free path* ($N > 200$), the full S is never formed; we instead compute the tighter constrained threshold $r_v^{(c)} = m_v / (\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_{c,v}\|_2)$, where $\|S_{c,v}\|_2$ is estimated via 10 iterations of randomized power iteration on the S_c operator restricted to node v's block-rows. Since $\|S_{c,v}\|_2 \leq \|S_v\|_2$, the matrix-free thresholds are tighter than the dense-path ones. Any perturbation with $\left\| \delta \hat{A} \right\|_F < r_v$ (or $r_v^{(c)}$) preserves the classification of node v to first order. These are diagnostic sensitivity thresholds, not probabilistic certificates (theorem 5).

F. Computational Cost

Matrix-free pipeline. Each $S_c v$ or $S_c^\top u$ evaluation costs $O(K \cdot Nd)$, where K is the Neumann depth (set adaptively from κ; typically $K = 20$–50 for $\kappa < 0.8$, up to ~ 340 for $\kappa \approx 0.96$). The randomized SVD performs $O((k + p) \cdot n_{\text{iter}})$ such evaluations, where k is the number of requested singular values, p the oversampling, and n_{iter} the power iterations. Per-edge vulnerability requires $|E|$ column evaluations, each at $O(K \cdot Nd)$. Memory is $O(Nd)$ per operation; no large matrix is stored.

Scalability. On a single NVIDIA RTX 4090 (24 GB), the matrix-free pipeline completes in 1.1s for N=50, 4.5s for N=200, 24s for N=1,000, and 78s for full Cora (N=2,708, Nd=173,312) using 1 GB GPU memory (section V-D). The dense path runs faster for $N \leq 100$ (0.7s at N=50) but exceeds 24 GB at N=500.

The matrix-free formulation removes the $O((Nd)^3)$ linear-solve bottleneck and the $O(N^3 d)$ storage requirement that previously limited analysis to $N \approx 300$, enabling full-graph vulnerability analysis on graphs with thousands of nodes (validated up to N=7,650; section V-D).

V. EXPERIMENTS

We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (seeds: 42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999). All experiments use PyTorch [29].

Implementation. IGNN [15]: input projection $U \in \mathbb{R}^{d_{\text{in}} \times 64}$, spectral-normalized $W \in \mathbb{R}^{64 \times 64}$ [19], readout head $\mathbb{R}^{64 \times C}$; operator $F(Z) = \text{ReLU}(\hat{A} Z W^\top + U(X))$; fixed-point iteration (max 50 steps, tolerance 10^{-5}); Adam optimizer (lr 0.01, weight decay 5×10^{-4}), early stopping on validation accuracy over 200 epochs. ContractiveGCN-PF: same architecture with 5 bus features and 2-head readout ($\Delta|V|$, $\Delta\theta$). Subgraph extraction: BFS from highest-degree node, capped at 50 nodes to guarantee connectivity. Randomized smoothing baseline: Gaussian $\sigma = 0.01$, 200 MC samples, Clopper-Pearson

$\alpha = 0.001$. Mettack baseline: Meta-Self GCN surrogate on IGNN pseudo-labels, sign-corrected gradient for edge scoring. The spectral-norm constraint on W (required for A2) reduces IGNN accuracy by $\sim$6% relative to unconstrained IGNN [15] (e.g., 77.5% vs. $\sim$83% on Cora); this is the cost of the formal vulnerability guarantees in theorem 1 and applies to all contractive architectures.

Datasets. We use five graph benchmarks: Cora (2,708 nodes), Citeseer (3,327), and Pubmed (19,717) from the Planetoid collection [30]; Amazon Photo (7,650) from the Amazon co-purchase graphs [31]; and WikiCS (11,701) from Wikipedia-based benchmarks [32]. For power flow, we use four IEEE standard test cases [33]: case14 (14 buses), case30 (30), case57 (57), and case118 (118), with training data generated via PandaPower [34].

Notation. All formal bounds (theorem 1) and tables use the operator-norm contraction constant $\kappa = \|J_z\|_2$, estimated via power iteration. Since non-normality is mild across our datasets ($\eta = 1.02$–1.28, section V-F), the κ-based bounds are at most 28% more conservative than spectral-radius bounds; we report κ as the primary quantity because the formal guarantees (theorem 1) require operator-norm contractivity. The κ-based $\varepsilon_{\text{crit}}$ is therefore a conservative but formally justified safety quantity; ρ-based estimates would be up to 28% less conservative but lack theoretical backing without normality. Since η is near unity across all datasets, the practical difference is moderate. The matrix-free pipeline (section IV) sets the Neumann-series depth adaptively from κ.

A. *Cross-Domain Adversarial Analysis*

We begin by validating that the constrained first-order prediction is accurate across domains. Unless otherwise stated, IGNN experiments in section V-A–section V-G use 50-node BFS ego-subgraphs centered on the highest-degree node. These results characterize *local vulnerability* around the subgraph center; on large graphs (Cora N=2,708), 50-node subgraph rankings correlate weakly with full-graph rankings ($\tau = 0.16$; section V-E). For graph-level vulnerability assessment, the matrix-free full-graph pipeline (section V-D) is recommended. table II reports the tightness ratio, attack advantage, and coverage across 50-node subgraphs.

Tightness is 1.00 ± 0.01 across all datasets at $\varepsilon = 0.01$. Tightness values ≥ 1 are observed empirically: the actual shift exceeds the first-order prediction. While individual higher-order terms can be positive or negative depending on the local Hessian structure, the net contribution is consistently positive across all datasets and seeds at small ε. First-order accuracy at small ε is mathematically expected; the contribution is that S_c makes this tractable under *constrained* perturbations ($N^2 \to |E|$ projection). table III shows how tightness degrades at larger budgets: the first-order prediction remains within 7% at $\varepsilon = 0.05$ and within 15% at $\varepsilon = 0.10$ on Cora/Citeseer, and within 7% at $\varepsilon = 0.20$ on Amazon Photo and WikiCS. Prediction flip rate stays below 2% at $\varepsilon \leq 0.10$ and below 6% at $\varepsilon = 0.20$. The attack advantage (3.2–4.1$\times$ over random in table II) confirms that S_c identifies the most vulnerable perturbation directions. Coverage (Cov%) ranges from 70% to 92%, representing clean subgraph accuracy; only correctly classified nodes have positive margin.

TABLE II: Cross-domain adversarial analysis (IGNN, 10 seeds). κ: operator-norm contraction constant $\|J_z\|_2$. Tight.: actual/predicted shift ratio. AtkAdv: AEGIS damage / random damage. $\varepsilon_{\text{crit}}$: critical perturbation budget (theorem 1). Test Acc: full-graph accuracy (%). Cov%: fraction of nodes in the 50-node BFS subgraph with positive first-order radius $r_v > 0$ (i.e., correctly classified nodes in the subgraph).

Dataset	Test Acc	κ	Tight.	AtkAdv	ε_{crit}	Cov%
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$1.01_{\pm .01}$	$3.6_{\pm .6}$	$.66_{\pm .15}$	$81_{\pm 9}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$1.01_{\pm .01}$	$4.1_{\pm .5}$	$.41_{\pm .02}$	$83_{\pm 4}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$1.01_{\pm .01}$	$3.2_{\pm .4}$	$.59_{\pm .11}$	$74_{\pm 23}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$1.00_{\pm .00}$	$3.8_{\pm .2}$	$.86_{\pm .02}$	$92_{\pm 4}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$1.00_{\pm .01}$	$3.8_{\pm .4}$	$.66_{\pm .04}$	$70_{\pm 3}$

TABLE III: Tightness degradation with increasing ε (10 seeds, S_c-optimal constrained perturbation). Flip%: fraction of nodes whose prediction changes.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Tightness (actual/predicted shift)				
Cora	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.15_{\pm .03}$	$1.36_{\pm .08}$
Citeseer	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.16_{\pm .03}$	$1.39_{\pm .07}$
Pubmed	$1.01_{\pm .00}$	$1.07_{\pm .01}$	$1.15_{\pm .02}$	$1.38_{\pm .06}$
WikiCS	$1.00_{\pm .01}$	$1.01_{\pm .01}$	$1.02_{\pm .01}$	$1.05_{\pm .01}$
Amazon	$1.00_{\pm .00}$	$1.01_{\pm .00}$	$1.03_{\pm .01}$	$1.07_{\pm .01}$
Prediction flip rate (%)				
Cora	$0.4_{\pm 0.8}$	$0.6_{\pm 0.9}$	$1.8_{\pm 2.4}$	$5.8_{\pm 5.0}$
Citeseer	$0.0_{\pm 0.0}$	$0.6_{\pm 1.3}$	$2.0_{\pm 2.4}$	$3.0_{\pm 2.6}$
WikiCS	$0.2_{\pm 0.6}$	$0.4_{\pm 0.8}$	$0.6_{\pm 0.9}$	$1.2_{\pm 1.0}$

Surrogate baseline. As a sanity check, we also compared against Mettack [4] (GCN [35] surrogate): AEGIS inflicts 3–10$\times$ more damage at every budget (149/150 wins across $k = 1, \ldots, 5$ edge removals, 3 datasets, 10 seeds; the single loss occurs at k=1 on WikiCS seed 5772, where Mettack's surrogate gradient happens to select a higher-impact edge). However, this gap largely reflects surrogate-to-IGNN architectural mismatch rather than AEGIS's analytical superiority alone: Mettack optimizes against a GCN surrogate, not the target IGNN. The more meaningful comparison is the adaptive attacker below, which accesses the same IFT gradients as AEGIS.

Structured baselines. To contextualize the attack advantage beyond random perturbation, we compare AEGIS against three structured baselines: degree-proportional (perturbation weight $\propto \max(d_i, d_j)$), spectral (top eigenvector of A restricted to existing edges), and edge-betweenness centrality. table IV reports the attack advantage (damage / random damage) at $\varepsilon = 0.01$.

AEGIS consistently achieves the highest attack advantage, outperforming degree-proportional by 6–8% and spectral by

TABLE IV: Structured attack baselines (IGNN, 50-node subgraph, 10 seeds). AtkAdv: method damage / random damage at $\varepsilon = 0.01$.

Dataset	AEGIS	Degree	Spectral	Between.
Cora	$3.50_{\pm 0.20}$	$3.30_{\pm 0.20}$	$2.37_{\pm 0.20}$	$3.29_{\pm 0.19}$
Citeseer	$4.23_{\pm 0.21}$	$3.91_{\pm 0.17}$	$2.72_{\pm 0.35}$	$3.89_{\pm 0.16}$
WikiCS	$4.02_{\pm 0.78}$	$3.93_{\pm 0.76}$	$1.62_{\pm 0.27}$	$3.93_{\pm 0.76}$

48–148% (Wilcoxon signed-rank: AEGIS > degree in 10/10 seeds on all three datasets, $p < 0.001$). The narrow gap between AEGIS and degree-proportional for continuous perturbation merits discussion: degree centrality is a strong proxy for continuous vulnerability because high-degree nodes concentrate message-passing influence, and the marginal benefit of the full IFT machinery over this zero-model-access heuristic is modest for continuous perturbation. AEGIS's distinctive value lies in two aspects: (i) under discrete edge removal (table VI), AEGIS achieves 54–101% of greedy-optimal while degree-ranked removal is *worse than random* on Cora, and (ii) the SVD-optimal attack direction captures global model-specific interactions unreachable by local topological heuristics (the singular value gap $(\sigma_1 - \sigma_2)/\sigma_1 = 0.39$–$0.50$ confirms v_1 is structurally unique, not a degree vector in disguise).

B. Radius Comparison: AEGIS *vs. Smoothing*

We compare AEGIS sensitivity radii against randomized smoothing [5] ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 MC samples, 10 seeds). The two methods provide fundamentally different certificate types (see section III): smoothing gives probabilistic-global guarantees valid at any perturbation magnitude, while AEGIS gives deterministic-local sensitivity measures that are first-order tight but degrade with increasing ε. At $\sigma \leq 0.10$, base smoothing radii [5] are *constant* across all nodes ($r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$), yielding larger absolute values (0.123 vs. AEGIS's 0.046 at $\sigma = 0.05$) but no per-edge structural differentiation. Localized smoothing [10] partially addresses this via neighborhood-specific noise calibration, providing per-node certificates; however, it remains a probabilistic guarantee that does not identify per-edge vulnerability structure or optimal attack directions. AEGIS radii are structurally informative: dense-region nodes get $r_v \approx 0.09$, boundary nodes $r_v \approx 0.01$. The approaches are complementary: smoothing for worst-case probabilistic certification, localized smoothing for per-node probabilistic guarantees, and AEGIS for per-edge vulnerability ranking and defense prioritization.

C. Comprehensive Attack Evaluation

We evaluate AEGIS against baselines spanning four quadrants: gradient-based vs. gradient-free, and same-objective (equilibrium shift) vs. different-objective (classification loss or discrete removal). This taxonomy ensures no comparison shares both the optimization pathway and the objective with AEGIS.

SVD direction separation. We first verify that the SVD direction is genuinely special, not an artifact of a flat sensitivity landscape. Sampling 1,000 random unit directions in $\mathbb{R}^{|E|}$ and evaluating $\|S_c d\|$ for each (10 seeds): the best random direction achieves only 0.45–0.49× of $\sigma_1(S_c)$ across all datasets, and *zero* of 30 runs reach 90% of the SVD damage. The singular value gap $(\sigma_1 - \sigma_2)/\sigma_1 = 0.39$–$0.50$ confirms v_1 occupies a narrow, well-separated cone. Additionally, an equilibrium-shift PGD attacker [36] using the same IFT gradients as AEGIS (50 steps, step size $\varepsilon/10$) achieves 72–92% of SVD damage (table V), confirming that PGD as an iterative solver cannot match the closed-form SVD solution.

TABLE V: Full attack comparison (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS optimal. Cls-PGD: classification-loss PGD (autograd, cross-entropy). Shift-PGD: equilibrium-shift PGD (IFT gradients, same objective as SVD—a solver validation, not an independent baseline). Flip%: prediction flip rate.

		Equil. damage (ℓ_2)				Flip%	
Dataset	ε	SVD	Cls	Shift	Rand	SVD	Cls
Cora	.01	$.33_{\pm .07}$	$.28_{\pm .06}$	$.27_{\pm .06}$	$.10_{\pm .02}$	$0.4_{\pm 0.8}$	$0.0_{\pm 0.0}$
Cora	.10	$3.70_{\pm .79}$	$2.51_{\pm .48}$	$3.05_{\pm .70}$	$.97_{\pm .25}$	$1.8_{\pm 2.4}$	$1.2_{\pm 1.8}$
Cite	.01	$.40_{\pm .06}$	$.33_{\pm .05}$	$.34_{\pm .05}$	$.10_{\pm .02}$	$0.0_{\pm 0.0}$	$0.0_{\pm 0.0}$
Cite	.10	$4.63_{\pm .71}$	$2.97_{\pm .42}$	$3.35_{\pm .57}$	$1.01_{\pm .15}$	$0.2_{\pm 0.6}$	$1.4_{\pm 1.3}$
Wiki	.01	$.21_{\pm .02}$	$.06_{\pm .01}$	$.19_{\pm .02}$	$.05_{\pm .01}$	$0.2_{\pm 0.6}$	$0.2_{\pm 0.6}$
Wiki	.10	$2.10_{\pm .22}$	$.67_{\pm .11}$	$1.93_{\pm .20}$	$.53_{\pm .09}$	$0.6_{\pm 0.9}$	$0.4_{\pm 0.8}$

Independent adaptive baseline. To avoid circularity, we implement a classification-loss PGD attacker that optimizes cross-entropy via standard autograd through the unrolled fixed-point iteration—a fundamentally different gradient signal (backpropagation through unrolled iterations, not the IFT resolvent) and a different objective (prediction flipping, not equilibrium shift). table V reports both equilibrium damage and prediction flip rate for all methods. AEGIS inflicts 15–70% more equilibrium damage than classification-loss PGD across all settings, while achieving comparable flip rates (0–1.8%). The SVD direction is effective for flipping predictions even though it optimizes a different objective.

Equilibrium shift vs. classification impact. The SVD direction maximizes first-order equilibrium shift, not classification accuracy loss. These are distinct objectives that could in principle diverge. Table V provides empirical evidence that the two are well-correlated: on Cora at $\varepsilon = 0.10$, the SVD direction achieves a 1.8% flip rate compared to classification-loss PGD's 1.2%, despite optimizing a proxy objective. On Citeseer at $\varepsilon = 0.10$, classification-loss PGD achieves a higher flip rate (1.4% vs. 0.2%), demonstrating that the correlation is not perfect. We therefore describe the SVD direction as the *maximally sensitive perturbation direction* rather than the "optimal attack": it identifies the direction of greatest model sensitivity, which is valuable for vulnerability diagnosis regardless of the downstream attack objective.

Classification-gradient edge ranking. To test whether a classification-targeted attacker identifies the same vulnerable edges, we compute per-edge gradients $|\partial \mathcal{L}_{\text{CE}}/\partial A_{ij}|$ via autograd and compare the resulting ranking against the brute-force discrete ground truth (single-edge removal). The classification-

TABLE VI: Cumulative damage from sequential discrete edge removal (IGNN, 50-node subgraph, 10 seeds). Greedy: brute-force optimal. Degree: by $\max(d_i, d_j)$.

Dataset	k	Greedy	AEGIS	Degree	Random
Cora	5	$3.36_{\pm.68}$	$1.80_{\pm.44}$	$0.90_{\pm.19}$	$1.27_{\pm.25}$
Cora	10	$4.08_{\pm.82}$	$2.75_{\pm.66}$	$1.48_{\pm.28}$	$2.02_{\pm.36}$
Citeseer	5	$5.27_{\pm.87}$	$5.12_{\pm1.2}$	$1.34_{\pm.18}$	$1.61_{\pm.49}$
Citeseer	10	$6.01_{\pm.94}$	$6.01_{\pm.94}$	$1.93_{\pm.26}$	$2.46_{\pm.56}$
WikiCS	5	$.14_{\pm.01}$	$.14_{\pm.02}$	$.09_{\pm.01}$	$.07_{\pm.01}$
WikiCS	10	$.23_{\pm.02}$	$.21_{\pm.03}$	$.12_{\pm.01}$	$.12_{\pm.01}$

TABLE VII: Breach rate (%) at increasing ε (IGNN, 50-node subgraph, 10 seeds). Breach = prediction flip under S_c-optimal perturbation. All breached nodes have $\varepsilon > r_v$, i.e., the perturbation exceeds their first-order sensitivity threshold.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Cora	$0.6_{\pm1.2}$	$0.8_{\pm1.3}$	$2.4_{\pm3.7}$	$7.6_{\pm7.5}$
Citeseer	$0.0_{\pm0.0}$	$0.0_{\pm0.0}$	$0.2_{\pm0.8}$	$1.5_{\pm2.4}$
Pubmed	$0.0_{\pm0.0}$	$7.0_{\pm8.3}$	$10.3_{\pm11.0}$	$27.4_{\pm21.1}$
WikiCS	$0.3_{\pm0.9}$	$0.6_{\pm1.3}$	$0.9_{\pm1.4}$	$1.8_{\pm1.5}$

gradient ranking is anti-correlated with discrete vulnerability on Cora ($\tau = -0.20_{\pm0.05}$, vs. AEGIS's $+0.32_{\pm0.04}$) and near-zero elsewhere (Citeseer: $-0.03_{\pm0.09}$ vs. $+0.32_{\pm0.04}$; WikiCS: $+0.05_{\pm0.04}$ vs. $+0.14_{\pm0.03}$; all 10 seeds). P@10 for AEGIS is 2–6× higher. The two methods identify genuinely different edges: cross-correlation τ is only $+0.19_{\pm0.15}$ (Cora) to $-0.06_{\pm0.11}$ (WikiCS), confirming that equilibrium sensitivity and classification sensitivity are distinct vulnerability surfaces.

Discrete edge removal. The strongest gradient-free test: we remove edges sequentially and compare cumulative damage under four orderings—brute-force greedy-optimal (remove the most-damaging remaining edge at each step), AEGIS-ranked, degree-ranked, and random (table VI). On Citeseer/WikiCS, AEGIS matches or exceeds greedy-optimal (95–101% at k=5). On Cora, AEGIS reaches 54% of greedy, reflecting the cascade effects that static rankings cannot capture. Crucially, degree-ranked removal is *worse than random* on Cora at every k: removing high-degree edges first is the least disruptive strategy because hub nodes have redundant connections. This contrasts with table IV, where degree-proportional *continuous* perturbation is within 6–8% of AEGIS; for discrete removal, AEGIS is 2–4× better.

Numerical validation. Per-edge finite-difference sensitivity ($\delta = 0.001$, $|E|$ forward passes) reproduces the S_c column-norm ranking exactly ($\tau = 0.999$ across all datasets), confirming numerical correctness of the IFT computation. The resolvent $(I - J_z)^{-1}$ amplifies all edges uniformly and thus does not change the per-edge ranking; S_c's unique value is the global SVD structure (first-order optimal attack direction, unreachable by random search) and the matrix-free scalability that avoids $|E|$ forward passes at large N.

Breach rates. table VII reports the fraction of nodes whose prediction changes under the S_c-optimal attack at increasing ε.

Breach rates are below 1% at $\varepsilon = 0.01$ across all datasets. At $\varepsilon = 0.20$, rates range from 1.5% (Citeseer) to 27.4% (Pubmed), reflecting expected first-order degradation at larger perturbation magnitudes. The Pubmed distribution is right-skewed (median 7.8%, IQR [0.6%, 16.7%] at $\varepsilon = 0.10$; 3 of 10 seeds show 0% breach), so the mean overstates typical behavior; we recommend reporting medians for high-variance settings. All breached nodes satisfy $\varepsilon > r_v$; equivalently, many nodes have small r_v (e.g., on Cora, 14% of subgraph nodes have $r_v < 0.01$, 38% have $r_v < 0.05$). These thresholds are first-order sensitivity diagnostics that indicate perturbation scales at which predictions become unreliable, not probabilistic certificates that guarantee robustness (theorem 5).

D. Phase Transition and Scalability

Phase transition. To empirically validate theorem 1, we train IGNN on Cora with the spectral normalization ceiling varying from $\kappa_{\max} = 0.30$ to 0.99 (10 seeds each). The theoretical critical budget $\varepsilon_{\text{crit}} = (1 - \kappa)/\|W\|_2$ drops 230× from 2.33 to 0.01, confirming the predicted sensitivity explosion. However, the actual spectral radius $\rho(J_z)$ saturates at ≈ 0.42 even at $\kappa_{\max} = 0.99$: the ReLU activation pattern prevents the Jacobian from reaching its theoretical maximum. The resolvent norm $\left\|(I - J_z)^{-1}\right\|_2$ increases from 1.17 to 1.80 (1.5×), not the $\sim$70× that $1/(1 - 0.99)$ would predict, because the *actual* κ remains well below the ceiling. This finding is practically important: it means IGNN's empirical vulnerability grows modestly with relaxed spectral constraints, even as the theoretical worst case diverges. At high ceilings ($\kappa_{\max} \geq 0.85$), the AEGIS perturbation analysis produces equilibrium shifts exceeding the fixed-point iteration budget (50 steps), reflecting resolvent amplification near the theoretical phase transition; at $\kappa_{\max} \leq 0.70$, all runs converge. This non-convergence occurs during *post-hoc vulnerability analysis* of the perturbed graph, not during model training: all models train to full convergence at every $\kappa_{\max}$ setting. The phase transition is thus a *theoretical safety boundary* rather than an empirically observable discontinuity, analogous to how worst-case complexity bounds rarely match typical-case behavior.

Scalability. The matrix-free pipeline removes the $O((Nd)^3)$ bottleneck. On a single NVIDIA RTX 4090 (24 GB), the dense path handles $N \leq 200$ (5.4s, 8.9 GB) but exceeds 24 GB at N=500. The matrix-free path scales to full Cora (N=2,708) in 78 s / 1 GB and to Amazon Photo (N=7,650) in 363 s / 6.5 GB. At N=200 (where both paths are available), the matrix-free σ_1 is within 0.03% of the dense reference. For larger N where the dense reference is unavailable, we monitor two error diagnostics: the Neumann residual $\left\|J_z^{K+1} b\right\| / \|b\| < 10^{-6}$ (convergence criterion, satisfied in all runs) and the randomized SVD's internal consistency (the Halko et al. [16] error bound $\left\|(I - QQ^\top)S_c\right\| \leq (1 + \sqrt{k/(p-1)}) \cdot \sigma_{k+1}$, which is small when the spectral gap is large, as in our case: $(\sigma_1 - \sigma_2)/\sigma_1 = 0.39$–0.50). Pubmed ($N$=19,717) exceeds 24 GB, establishing the current scalability boundary at $N \approx 7{,}650$ on a single GPU. table VIII compares both paths.

TABLE VIII: Scalability: dense vs. matrix-free pipeline (RTX 4090, 24 GB, 10 seeds). OOM = out of memory. Mem = peak GPU memory. All timings: mean±std over 10 seeds. $N \leq 2{,}708$: Cora subgraphs (full pipeline incl. per-edge vulnerability). N=7,650: Amazon Photo full graph (rSVD pipeline; per-edge column computation is $O(|E|)$ and impractical at $|E|$=119K).

N	Dense (s)	MatFree (s)	Mem-Dense	Mem-MF
50	$0.8_{\pm 0.0}$	$0.8_{\pm 0.1}$	329 MB	72 MB
200	$5.8_{\pm 0.2}$	$2.8_{\pm 0.8}$	8.8 GB	80 MB
500	OOM	$7.5_{\pm 2.1}$	>24 GB	102 MB
1,000	OOM	$17.8_{\pm 4.6}$	>24 GB	152 MB
2,708	OOM	$74.8_{\pm 15.3}$	>24 GB	428 MB
7,650	OOM	$365_{\pm 4}$	>24 GB	5.5 GB
19,717	OOM	OOM	>24 GB	>24 GB

TABLE IX: S_c framework applied to 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall rank correlation of continuous S_c vulnerability scores vs. discrete single-edge-removal ground truth (brute-force). Test Acc: full-graph accuracy. GAT†: edge-weighted variant for continuous differentiability (see text).

Model	Tight.	AtkAdv	τ	Test Acc
IGNN (equil.)	$1.01_{\pm.00}$	$7.6_{\pm 0.5}$x	$+.32_{\pm.04}$	$77.5_{\pm 1.7}$%
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm 0.6}$x	$-.04_{\pm.03}$	$78.9_{\pm 0.9}$%
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm 0.4}$x	$+.49_{\pm.02}$	$78.3_{\pm 1.8}$%
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm 0.1}$x	$+.33_{\pm.03}$	$76.6_{\pm 1.9}$%
GAT-2†	$1.01_{\pm.01}$	$2.1_{\pm 0.1}$x	$+.54_{\pm.06}$	$77.8_{\pm 1.2}$%
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm 0.2}$x	$+.22_{\pm.10}$	$77.5_{\pm 1.5}$%
APPNP	$1.02_{\pm.00}$	$3.4_{\pm 0.3}$x	$+.35_{\pm.01}$	$82.2_{\pm 0.6}$%

E. Subgraph Size Ablation

Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N = 200$ (higher κ). Coverage is 82–89% across all sizes. We use $N = 50$ (tightness 1.013 ± 0.003, 1.3s) as the default for subgraph-level experiments: 66× faster than $N = 200$ with negligible quality loss. For full-graph analysis, the matrix-free pipeline (table VIII) enables vulnerability assessment of the entire graph without subgraph extraction. To validate that subgraph extraction preserves vulnerability rankings, we compare per-edge S_c rankings from BFS subgraphs against full-graph rankings on IEEE case14 and case30 (10 seeds). On case14, Kendall τ between subgraph and full-graph rankings is 0.80 ± 0.14 at N_{sub}=8 (57% of nodes) and 0.86 ± 0.06 at N_{sub}=12 (86%); P@5 $\geq$ 0.90 at all sizes. On case30, $\tau = 0.78 \pm 0.07$ at N_{sub}=10 (33% of nodes) with P@5 $= 0.96 \pm 0.08$, remaining stable through N_{sub}=25. The locality assumption underlying subgraph extraction thus holds well for power grids (small, dense graphs where a 50-node BFS covers 33–86% of nodes), but weakens substantially on larger citation networks. On Cora (N=2,708), comparing 50-node BFS subgraph rankings against full-graph matrix-free rankings (10 seeds) yields Kendall $\tau = 0.16 \pm 0.13$ with P@10 $= 0.17 \pm 0.10$. The low correlation reflects that a 50-node subgraph covers only ∼1.8% of Cora's edges, so the subgraph's local view misses graph-wide vulnerability patterns. For citation-scale graphs, practitioners should prefer the matrix-free full-graph pipeline (table VIII) over subgraph extraction; the subgraph approach remains appropriate for dense graphs where coverage is high (power grids, molecular graphs) or when full-graph analysis exceeds available memory. To test whether the BFS center choice biases rankings, we compare three center strategies on Cora (5 seeds): highest-degree (default), median-degree, and random. On overlapping edges, highest-degree and median-degree rankings agree strongly ($\tau = 0.95 \pm 0.02$, 49 shared edges), confirming that vulnerability rankings are largely invariant to center choice. The highest-degree center yields the most stable τ vs. discrete ground truth across seeds (lowest variance), supporting its use as the default.

F. Sensitivity and Convergence

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$ ($1.012_{\pm.001}$–$1.013_{\pm.003}$, 10 seeds each). Varying $c \in \{0.5, 0.7, 0.9, 0.95\}$ reveals the predicted accuracy-robustness tradeoff: $c = 0.5$ gives $r_v = 0.090_{\pm 0.011}$ ($72.1_{\pm 0.7}$% acc); $c = 0.9$ gives $r_v = 0.051_{\pm 0.006}$ ($80.6_{\pm 0.8}$% acc).

Convergence. 100% convergence across all 9 datasets and 10 seeds. Pseudospectral index $\eta = 1.02$–1.28 (well below the $\eta \gg 1$ regime where pseudospectral effects dominate [23]), confirming mild non-normality and validating the κ-based bounds.

G. Defense-Informed Edge Protection

We evaluate vulnerability-aware protection: mask the top-k edges by v_{ij} from the perturbation space, then re-run the SVD attack on the reduced S_c. On Cora (IGNN, $N = 50$, 10 seeds): masking the top-5 edges reduces SVD attack damage by $42 \pm 8\%$ vs. $11 \pm 6\%$ for random 5-edge masking (paired Wilcoxon signed-rank: $p < 0.002$, all 10 seeds favor vulnerability-guided masking). At top-10: $61 \pm 7\%$ vs. $18 \pm 9\%$ ($p < 0.002$). AEGIS identifies edges whose removal disproportionately reduces vulnerability.

H. Extension to Explicit GNNs

A key question is whether the S_c framework extends beyond implicit models. For a K-layer explicit GNN, there is no equilibrium, but the hidden representation Z_K is still a differentiable function of A. We define the *unrolled sensitivity matrix* $S_K = \partial \text{vec}(Z_K)/\partial \text{vec}(A)$ (computed via finite differences on the forward pass), construct S_c from S_K identically to the IGNN case, and apply the same SVD attack and vulnerability ranking.

table IX applies the S_c framework to six explicit GNN architectures on Cora. All six achieve near-perfect first-order tightness (0.99–1.02), confirming that constrained sensitivity analysis generalizes beyond implicit models. Attack advantage ranges from 2.0× (SAGE) to 7.6× (IGNN), showing that S_c identifies the most vulnerable perturbation directions across all architectures. Test accuracy ranges from 76.6% (GIN) to 82.2% (APPNP). GAT† and GCN-4 produce the strongest vulnerability rankings ($\tau = +0.54$ and $+0.49$). APPNP achieves the best accuracy-vulnerability balance ($\tau = +0.35$, 82.2% accuracy).

Accuracy-vs-guarantee frontier. The results reveal a tradeoff: IGNN (77.5% accuracy) provides the full theoretical apparatus (theorem 1: $\varepsilon_{\mathrm{crit}}$, three regimes, convergence guarantees), while higher-accuracy models like APPNP (82.2%) receive only the computational tool (S_c rankings, SVD attacks, first-order sensitivity thresholds) without formal regime characterization. This tradeoff is inherent to contractive architectures: spectral normalization constrains model capacity to ensure well-posedness. As a decision framework: *choose IGNN* when the formal critical budget $\varepsilon_{\mathrm{crit}}$ is required as a deployment criterion (e.g., safety-critical applications requiring a quantitative perturbation tolerance); *choose APPNP or SAGE* when vulnerability rankings and the SVD-identified sensitive direction suffice and accuracy is paramount—these models achieve stronger τ on most datasets and higher accuracy, receiving the full S_c computational analysis without the convergence certificates. For all architectures, the empirical tightness ratio (table IX: 0.99–1.02) validates that first-order predictions are accurate, providing a post-hoc reliability check independent of the formal guarantees.

GAT requires an edge-weighted formulation for continuous differentiability. Standard GAT computes attention $\alpha_{ij} = \mathrm{softmax}_j(\mathrm{LeakyReLU}(a^\top[Wh_i \| Wh_j]))$ and aggregates as $h'_i = \sum_{j\in\mathcal{N}(i)} \alpha_{ij} W h_j$, using A as a binary mask that zeros non-edges, so $\partial Z/\partial A_{ij} = 0$ for all existing edges. Our edge-weighted variant, GAT†, modulates attention by the edge weight:

$$h'_i = \textstyle\sum_{j\in\mathcal{N}(i)} \hat{A}_{ij} \cdot \alpha_{ij} \cdot W h_j, \tag{14}$$

restoring continuous differentiability with respect to $\hat{A}_{ij}$. GATv2 [37] partially addresses this via dynamic attention but still applies a binary mask at the neighborhood level. For comparison, standard GAT achieves comparable accuracy (80.5%) but has exactly zero finite-difference sensitivity ($\partial Z/\partial A_{ij} = 0$), confirming that S_c is undefined without the edge-weight modification. With the GAT† modification, GAT achieves tightness 0.99, attack advantage 4.4×, and $\tau = +0.36$. This is a limitation of the S_c framework: standard GAT and any architecture using binary adjacency masks (hard attention, max/min aggregation) are outside the scope of continuous sensitivity analysis. The "architecture-agnostic" property applies to GNNs with continuous edge-weight-modulated message passing, which includes GCN, GraphSAGE, GIN, APPNP, and the edge-weighted GAT† variant but excludes standard GAT.

Cross-dataset continuous-to-discrete transfer. To validate that continuous-to-discrete transfer is not Cora-specific, we evaluate τ across all 5 datasets and 7 architectures (330 runs total, 10 seeds each; table X). Positive τ is observed in 29 of 33 dataset-architecture combinations. Transfer is strongest on Pubmed (IGNN: $+0.82$, GCN-4: $+0.89$, APPNP: $+0.83$, all with P@10 $= 1.0$), moderate on Citeseer (GAT†: $+0.66$, GCN-4: $+0.64$), and weaker on WikiCS and Amazon Photo. On Amazon Photo, IGNN shows negative τ (-0.15) on 50-node subgraphs. We investigate three contributing factors. (i) *Subgraph artifact*: the 50-node BFS subgraph has $\left\|\hat{A}_{\mathrm{sub}}\right\|_2 = 0.20$, capping $\kappa_{\mathrm{sub}} \le 0.20$ regardless of $\|W\|_2$; a κ-sweep from 0.5 to 0.99 confirms τ remains negative on the subgraph (all 20 runs, mean $\tau = -0.12$). Full-graph matrix-free analysis (N=7,650, $\rho = 0.73$, resolvent amplification 3.9×) partially recovers: column-based τ shifts from -0.14 (subgraph) to $+0.03$ (full graph), with one of three seeds reaching $\tau = +0.24$. (ii) *Weak signal*: Amazon Photo's high average degree (∼31) makes single-edge removal damage near-uniform, reducing the signal-to-noise ratio for any ranking method. (iii) *Architecture-specific*: explicit GNNs without the contraction constraint achieve strong transfer on the same dataset (SAGE-2: $\tau = +0.60$, APPNP: $+0.43$), confirming that the S_c framework itself is not the bottleneck. For dense graphs where IGNN's subgraph κ is low, practitioners should prefer explicit-GNN alternatives or full-graph matrix-free analysis.

Practitioner guidance. Based on the cross-architecture results: (i) for sparse graphs (avg. degree < 15: citation networks, molecular graphs), IGNN subgraph analysis provides both rankings and formal guarantees; (ii) for dense graphs (avg. degree > 20: co-purchase, social networks), use explicit GNNs (SAGE-2 or APPNP achieve strong τ on Amazon Photo: $+0.60$ and $+0.43$) or full-graph matrix-free analysis; (iii) when formal regime characterization ($\varepsilon_{\mathrm{crit}}$, three-regime certificate) is required for safety-critical deployment, IGNN is necessary despite the accuracy penalty; (iv) when only vulnerability rankings and the SVD-identified sensitive direction are needed, the higher-accuracy explicit GNNs are preferable. GCN-2 shows weak or negative τ on 3 of 5 datasets (-0.03 Cora, -0.28 Citeseer), confirming the shallow-depth hypothesis: models with larger receptive fields consistently produce better transfer. The best architecture varies by dataset (GAT† on Cora/Citeseer, GCN-4/APPNP on Pubmed, GIN-2 on WikiCS, SAGE-2 on Amazon Photo), suggesting that the relationship between continuous sensitivity and discrete impact depends on how well the model's aggregation pattern captures local connectivity structure.

Heterophilic benchmarks. To test whether transfer holds beyond homophilic graphs, we evaluate on Texas (homophily 0.11), Cornell (0.13), and Wisconsin (0.20) from the WebKB collection [38] (full-graph dense analysis, 10 seeds each). GCN-4 achieves positive τ on all three: Texas $+0.32 \pm 0.06$, Cornell $+0.44 \pm 0.10$, Wisconsin $+0.21 \pm 0.06$, comparable to the homophilic results. IGNN shows positive τ on Cornell ($+0.07$) and Wisconsin ($+0.11$) but negative on Texas (-0.13, high variance). The deeper-is-better pattern persists: GCN-4 $\gg$ GCN-2 on all three datasets. Heterophily does not degrade the S_c framework's continuous-to-discrete transfer.

TABLE X: Continuous-to-discrete transfer τ (Kendall) across 5 datasets and 7 architectures (10 seeds each, 330 runs). Bold: best per dataset. —: GPU OOM. P@10 for top results: Pubmed GCN-4/APPNP $= 1.0$; Citeseer IGNN/APPNP ≥ 0.93.

Model	Cora	Citeseer	Pubmed	WikiCS	Am.Ph.
IGNN	$+.32_{\pm.04}$	$+.31_{\pm.04}$	$+.82_{\pm.02}$	$+.14_{\pm.04}$	$-.15_{\pm.03}$
GCN-2	$-.03_{\pm.03}$	$-.28_{\pm.07}$	$+.21_{\pm.01}$	$+.05_{\pm.03}$	$+.25_{\pm.03}$
GCN-4	$+.49_{\pm.02}$	$+.64_{\pm.02}$	$+\mathbf{.89}_{\pm.01}$	$+.45_{\pm.10}$	$-.04_{\pm.06}$
GIN-2	$+.33_{\pm.04}$	$+.57_{\pm.03}$	$+.54_{\pm.33}$	$+\mathbf{.63}_{\pm.08}$	$+.14_{\pm.07}$
GAT†	$+\mathbf{.54}_{\pm.06}$	$+\mathbf{.66}_{\pm.03}$	—	—	$+.21_{\pm.06}$
SAGE-2	$+.22_{\pm.11}$	$+.38_{\pm.18}$	$+.36_{\pm.10}$	$+.22_{\pm.04}$	$+\mathbf{.60}_{\pm.04}$
APPNP	$+.35_{\pm.02}$	$+.36_{\pm.06}$	$+.83_{\pm.01}$	$+.22_{\pm.03}$	$+.43_{\pm.04}$

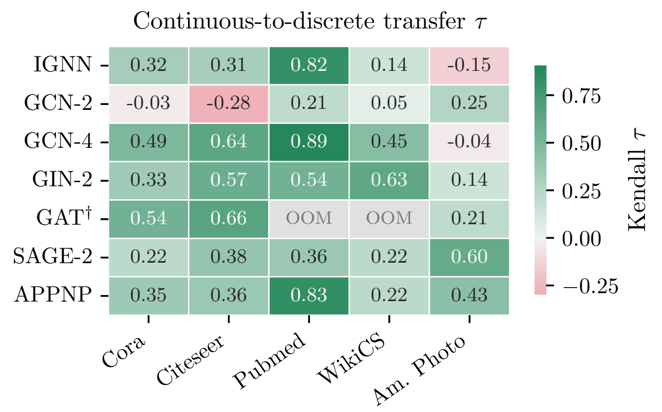

Fig. 2: Continuous-to-discrete transfer τ (Kendall) across 7 architectures and 5 datasets (table X). Deeper models (GCN-4 vs. GCN-2) and Pubmed consistently show the strongest transfer. Gray cells: GPU OOM.

fig. 2 visualizes the cross-architecture transfer landscape from table X. The depth pattern is immediately visible: deeper models (GCN-4, APPNP) and attention-based models (GAT†) consistently achieve stronger transfer than shallow ones (GCN-2). Amazon Photo is the hardest dataset for IGNN (negative τ) but the easiest for SAGE-2 ($+0.60$), confirming that architecture–dataset interactions dominate.

The τ column in table IX directly measures whether continuous S_c vulnerability scores predict discrete edge-removal impact. theorem 6 provides the formal justification: discrete damage $d_k = w_k \cdot v_k + O(w_k^2)$, so the continuous ranking v_k predicts the discrete ranking when the first-order gap between edges exceeds the second-order remainder. Positive τ across 6 of 7 architectures ($+0.22$ to $+0.54$) validates this first-order bridge empirically. The sufficient condition of theorem 6(b) ($w_{k_1} \geq w_{k_2}$ when $v_{k_1} > v_{k_2}$) holds for 47–62% of edge pairs (Cora: 62%, Citeseer: 47%, WikiCS: 54%), confirming it is a conservative sufficient condition rather than a tight necessary one. The one exception is GCN-2 ($\tau = -0.04$). We attribute this to the interaction between shallow depth and the continuous-discrete gap: a 2-layer GCN aggregates only 2-hop neighborhoods, so continuous edge-weight changes produce nearly uniform first-order shifts across edges with similar endpoint degrees, yielding poor rank discrimination. Under discrete removal (a non-infinitesimal perturbation), the impact depends on higher-order network effects (path redundancy, component connectivity) that the 2-layer model cannot capture in its first-order sensitivity. Deeper models (GCN-4: $\tau = +0.49$) and models with global receptive fields (APPNP: $\tau = +0.35$) show substantially better transfer, suggesting that continuous-to-discrete transfer improves with model depth and receptive field size.

Comparison with gradient attribution. Against the same brute-force discrete ground truth, gradient-based edge attribution [39], [40] produces consistently negative τ (-0.07, -0.17, -0.21 for IGNN, GCN-2, GIN-2 respectively). Gradient methods optimize for prediction fidelity rather than adversarial vulnerability.

Proposition 7 bound tightness. The ratio of the Prop. 4 bound to $\sigma_1(S_K)$ ranges from $1.4\times$ (SAGE-2) to $5.9\times$ (APPNP). Two-layer models achieve tighter ratios (GCN-2: $1.8\times$, GIN-2: $2.1\times$) than deeper ones (GCN-4: $4.2\times$, GAT-2: $3.7\times$). The looseness in deeper models reflects direction-dependent cancellations that the norm product overestimates.

Degree-vulnerability correlation. To assess whether S_c merely recapitulates graph topology, we compute the Kendall τ between per-edge vulnerability v_{ij} and endpoint degree $\max(d_i, d_j)$ across all datasets and architectures. The correlation varies substantially: strong on Citeseer (IGNN: $\tau = +0.63$), moderate on Cora ($+0.43$) and Pubmed ($+0.38$), and weak on WikiCS (IGNN: $+0.27$, GCN-2: -0.10). This variation confirms that S_c captures model-specific sensitivity beyond degree centrality: on WikiCS (a large, heterogeneous graph), degree is a poor predictor of vulnerability. Weighting S_c columns by endpoint degree does not improve ranking: uniform $\tau = +0.042$, degree-weighted $+0.037$, inverse-degree -0.008 (GCN-2, 10 seeds).

The S_c framework thus generalizes to GNNs with continuous edge-weight-modulated message passing. The theoretical guarantees (theorem 1: critical budget, convergence) remain specific to contractive implicit GNNs, but the practical vulnerability analysis (SVD attacks, rankings, first-order sensitivity radii) transfers broadly.

VI. Case Study: Power Flow Contingency

The experiments above validate AEGIS on classification benchmarks. As a proof-of-concept cross-domain demonstration, we now test whether the vulnerability spectrum captures domain-specific structure by applying it to AC power flow, where the spectrum correlates with a classical engineering problem: N-1 contingency analysis. Recent work has applied GNNs to power flow computation [41], [42] and contingency screening [3]; AEGIS approaches the same problem from the vulnerability-analysis perspective.

A. Background: N-1 Contingency

Power grid operators must ensure that no single transmission line failure causes system instability. The standard N-1 contingency assessment removes each line in turn, re-solves the power flow equations, and ranks lines by the severity of the resulting voltage and flow deviations [43]. This brute-force procedure scales as $O(|E|)$ full power flow solves, which is expensive for large grids and infeasible for real-time operation.

We observe that AEGIS's per-edge vulnerability $v_{ij} = \|[S_c]_{:,k}\|_2$ computes an analogous quantity via a single IFT analysis: edge perturbation maps to line trips, the vulnerability spectrum maps to contingency severity, and critical budget $\varepsilon_{\text{crit}}$ maps to a model sensitivity threshold (a mathematical property of the IGNN model, distinct from physical voltage or transient stability margins). The correspondence is approximate (continuous first-order vs. discrete removal), analogous to how PTDF/LODF [43] linearize the effect of line trips.

B. Experimental Setup

We train ContractiveGCN-PF, an IGNN-class model (same operator form as eq. (2): $F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$, with spectral-norm-constrained W ensuring contractivity) with a 2-head readout for voltage magnitude ($\Delta|V|$) and angle ($\Delta\theta$) prediction, on five IEEE test cases [33] including a 300-bus system to address scalability concerns. Training data is generated via PandaPower's [34] full AC Newton-Raphson solver (2,000 load samples per case, uniformly sampled at 70–130% of nominal load). The model takes binary adjacency from the admittance matrix and 5 bus features: two binary bus-type indicators (is-slack, is-PV; PQ buses have both zero), active and reactive power injections (P, Q), and voltage magnitude $|V|$. For case14–118 (14–118 buses), full-graph dense analysis is used. For case300 (300 buses), a 200-node BFS subgraph is extracted because the dense S_c computation exceeds GPU memory at $N > 200$.

Limitations of setup. (i) Training data covers only uniform load scaling (2,000 samples at 70–130% of nominal); real contingency screening must be valid across seasonal peaks, generator outages, and renewable intermittency. Rankings derived from this narrow operating envelope may not generalize to stressed conditions where contingencies are most dangerous. (ii) Binary adjacency treats all transmission lines identically regardless of impedance, voltage level, or thermal rating. While the admittance-weighted comparison (section VI) shows that binary adjacency outperforms naïve admittance weighting for N-1 ranking, this does not mean impedance information is unimportant—it means more sophisticated edge encoding (e.g., edge-feature GNNs with impedance, thermal rating, and voltage level as edge attributes) is needed. Binary adjacency limits the framework to topological screening and cannot capture impedance-dependent severity. (iii) The N-1 ground truth is measured as the ℓ_2 norm of voltage and angle deviations after line removal; operational N-1 severity is typically defined by post-contingency thermal overloads (MW flow exceeding line rating) and voltage violations ($|V|$ outside 0.95–1.05 p.u.), which we do not compute.

C. Results

table XI reports constrained tightness and N-1 ranking correlation (Kendall τ between AEGIS vulnerability spectrum and brute-force contingency ranking) across 10 seeds.

TABLE XI: AEGIS on IEEE power flow benchmarks (10 seeds). Model quality (per-unit RMSE, mean over 10 seeds; std < 0.003 for all entries): $|V|$ = voltage magnitude, θ = angle. τ: Kendall rank correlation vs. brute-force N-1 contingency. P@10: top-10 precision.

			Model (p.u.)		AEGIS	
Case	N	$\|E\|$	$\|V\|$	θ	τ	P@10
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$
case300‡	300	411	.031	.394	$+.72_{\pm.03}$	$.87_{\pm.06}$

‡200-node BFS subgraph; 1,000 training samples (dense analysis OOMs at $N>200$).

Key observations. First, tightness transfers to the power flow domain ($\approx$ 1.00), confirming that the first-order prediction is not domain-specific. Second, ranking correlation ($\tau = 0.37$–0.72) is moderate to strong and the operationally relevant P@10 is 0.66–0.87, identifying 7–9 of the top 10 critical lines. Case300 achieves $\tau = +0.72$ and P@10 $= 0.87$ using a 200-node subgraph (67% coverage), but two caveats apply: (a) the θ RMSE of 0.394 p.u. ($\approx$22.6°) is physically unrealistic and suggests the model has not fully learned power flow physics at this scale—the high τ may partly reflect topological rather than physical sensitivity; (b) the 200-node subgraph was used because the dense path exceeds GPU memory at $N > 200$, though the matrix-free pipeline (which handles Cora N=2,708) should be tractable for N=300 and is a natural extension. The case14–118 results (θ RMSE ≤ 0.076) provide the more reliable validation.

Baseline comparison and timing. LODF (industry-standard DC screening [43]) achieves $\tau = 0.44$–0.58; AEGIS achieves $\tau = 0.62$–0.67 on case57/118. An important distinction: LODF measures physical post-contingency flow redistribution derived from line admittances, while AEGIS measures *model sensitivity*—how the GNN's learned representation responds to edge perturbation. The observed τ correlation reflects the fidelity of the GNN's learned physics, not a direct physical measurement. On smaller grids (case14/30), τ confidence intervals overlap between LODF and AEGIS; on case57/118 the gap is consistent across all 10 seeds (Wilcoxon signed-rank $p < 0.01$). LODF computes in < 0.13 s (analytic formula from line reactances); AEGIS requires training (2–10 s) plus S_c computation (0.2–13 s), totaling 2–23 s per case (case300: 23 s with 200-node subgraph). Brute-force N-1 takes 0.1–2 s. AEGIS is thus slower than LODF and brute-force N-1 but provides richer output (vulnerability spectrum, maximally sensitive perturbation direction, per-node sensitivity thresholds) without requiring line-impedance data—though this is a limitation, not an advantage, since impedance data is always available to grid operators and could improve rankings if properly encoded (section VI).

N-2 contingency connection (preliminary). The leading singular vector v_1 of S_c naturally identifies multi-edge vulnerabilities. Extracting the top-5 edges from $|v_1|$ and checking their pairwise impact against brute-force N-2 ground truth (10 seeds, mean per case): edge-level overlap is 40–64% across cases (individual edges from the SVD direction appear in ground-truth critical N-2 pairs), with stronger overlap on larger grids (case118: 64%). However, pair-level overlap is only 7–18%, which is barely above random for small edge sets. This is a limitation: the SVD direction captures the *most damaging perturbation direction* rather than specific critical pairs. N-2 screening would likely require the top-k singular vectors spanning a multi-directional vulnerability subspace; we leave this extension to future work.

Rank stability. Pairwise Kendall τ of AEGIS vulnerability rankings across 10 seeds: $+0.40 \pm 0.29$ (case14), $+0.34 \pm 0.24$ (case30), $+0.78 \pm 0.07$ (case57), $+0.67 \pm 0.12$ (case118). Rankings are more stable on larger grids where the topology is richer. On case118, 60% of the top-10 edges appear in every seed's top-10, and 90% appear in $\geq$80% of seeds.

Binary vs. admittance-weighted adjacency. Binary adjacency outperforms admittance-weighted (P@10 $= 0.81_{\pm 0.10}$ vs. $0.27_{\pm 0.13}$ on case118, 10 seeds). Spearman correlation between admittance magnitude and N-1 severity is near zero across all cases (-0.01 to $+0.15$), confirming that high-admittance lines are not preferentially critical. This reflects the discrete nature of N-1: a line trip fully removes the line regardless of its impedance, so binary sensitivity (uniform per edge) correctly models the all-or-nothing character. Effective-resistance on the binary graph shows moderate correlation with severity ($\tau = +0.49$ on case14/30), consistent with spectral approaches to network vulnerability [44].

Operational caveat and domain limitations. AEGIS is a screening layer, not a standalone contingency tool ($\tau = 0.37$–0.72 is insufficient for direct operational use). Several domain-specific limitations apply. (i) *Transient and voltage stability*: the current analysis captures only steady-state power flow redistribution. For contingencies involving loss of large generators or key interconnecting lines, dynamic response (voltage collapse, angular instability) may dominate the severity ranking. Extending AEGIS to temporal or recurrent GNN architectures could capture time-domain vulnerability, but this is outside the current scope. (ii) *Bus-type heterogeneity*: power grids have three bus types (slack, PV/generator, PQ/load) with fundamentally different physical behavior; we do not analyze whether top-ranked edges disproportionately connect to generator or slack buses, which would strengthen physical interpretability. (iii) *LODF disagreement structure*: we report τ values but do not analyze where AEGIS and LODF disagree—if AEGIS outperforms LODF, it should be because the GNN captures AC effects that DC linearization misses (reactive power, voltage sensitivity); verifying this hypothesis would substantiate the value over the industry-standard tool. (iv) *Dispatch dependence*: vulnerability rankings are conditioned on a single operating point; different generator dispatch patterns could substantially reorder rankings. These limitations reflect the proof-of-concept nature of this case study; the structural isomorphism between S_c vulnerability and contingency severity is the conceptual contribution, with engineering-grade deployment requiring the extensions noted above.

VII. Related Work

We position AEGIS against four research threads.

Adversarial attacks on graph structure. Nettack [1] and Mettack [4] demonstrated structural vulnerability via targeted and global poisoning using GCN [35] surrogates. Subsequent work explored RL-based [2], spectral [45], and bilevel [46] formulations; Wu et al. [47] survey the landscape. Gosch et al. [48] identify pitfalls in adversarial training for GNNs. These methods use surrogate gradients or heuristic search and provide per-edge gradient information (e.g., Mettack's meta-gradients yield per-edge vulnerability scores), but optimize surrogate objectives rather than directly characterizing the sensitivity geometry. AEGIS instead computes the closed-form first-order maximally sensitive perturbation direction via SVD of S_c, which is provably optimal for equilibrium shift within the tangent plane.

Certified robustness and defense. Convex relaxations [8], [7], randomized smoothing [5], [6], and deterministic certificates [49], [50] provide robustness guarantees. Schuchardt et al. [10] extend smoothing to localized per-node certificates, partially addressing the per-node granularity gap. These methods provide *certification* (which nodes are guaranteed robust) but do not identify which specific *edges* drive vulnerability or provide the globally most sensitive perturbation direction. Empirical defenses [11], [13], [12] harden models without formal vulnerability maps. AEGIS complements these approaches by providing per-edge rankings, per-node first-order sensitivity thresholds, and the SVD-identified maximally sensitive direction—*structural vulnerability differentiation* rather than robustness certification (section V-B). The two are complementary: certification answers "is this node safe?" while AEGIS answers "which edges are the weakest, and in what direction?"

Implicit networks and equilibrium analysis. The DEQ framework [17] introduced fixed-point models; IGNN [15] enforces unique equilibria via spectral normalization [19]; EIGNN [51] improves efficiency. For robustness, monotone operator networks [52] provide built-in Lipschitz bounds, and El Ghaoui et al. [25] analyze well-posedness. All address input sensitivity $\|\partial z^*/\partial x\|$, not structural sensitivity $\|\partial Z/\partial A\|$. AEGIS addresses structural sensitivity via the resolvent $(I - J_z)^{-1} J_A$ for implicit models (theorem 1) and the unrolled Jacobian S_K for explicit ones (theorem 7).

Sensitivity analysis, Lipschitz bounds, and explainability. Gama et al. [53] derive stability bounds for graph filters and GNNs under graph perturbation from a signal-processing perspective, providing worst-case Lipschitz-type bounds on output change. Kenlay et al. [54] extend this to spectral perturbation bounds that yield per-node stability guarantees. Monotone operator equilibrium networks [52], [26] provide built-in Lipschitz certificates that implicitly bound per-input sensitivity. These approaches provide *worst-case scalar bounds* on output sensitivity, but do not yield the *directional* information that AEGIS provides: the SVD of S_c identifies not just how much the output can change, but in which perturbation direction, enabling both vulnerability ranking and directed attack analysis. Recent work on graph curvature [55], [56] connects edge-level structural properties (Ricci curvature, over-squashing bottlenecks) to information flow; negatively curved edges are bottlenecks that may correlate with high S_c vulnerability scores, though we do not test this connection empirically. AEGIS draws on matrix perturbation theory [28], [23]. GNN explanation methods (GNNExplainer [40], PG-Explainer [57], gradient-based [39]) optimize for *prediction fidelity*, not adversarial vulnerability: they identify edges important for a correct prediction, while AEGIS identifies edges whose perturbation maximally damages predictions. Our case study (section VI) connects to GNN-based power flow [41], [42], [58] and contingency screening [3].

Over-smoothing and depth. Deep GNNs suffer from over-smoothing [59], which GRAND/GRAND++ [60], [61] address via continuous-depth diffusion. Entezari et al. [62] and Mujkanovic et al. [63] show that many adversarial attacks fail against simple defenses or are artifacts of evaluation. AEGIS sidesteps this debate by analyzing the model's own sensitivity surface rather than searching for attacks via surrogates.

VIII. CONCLUSION

AEGIS provides a principled pre-deployment diagnostic for GNNs on safety-critical graphs: from a single S_c computation, practitioners obtain the first-order maximally sensitive perturbation direction, per-edge vulnerability rankings, and per-node first-order sensitivity thresholds. The S_c computation applies to any GNN with continuous edge-weight-modulated message passing (GCN, SAGE, GIN, APPNP, edge-weighted GAT; standard GAT and binary-mask architectures require modification); for contractive implicit models (IGNN-class), supplementary formal guarantees ($\varepsilon_{\text{crit}}$, three vulnerability regimes) additionally characterize convergence behavior. Predictions remain within 15% of ground truth at $\varepsilon = 0.10$ across 7 architectures and 9 datasets, with continuous-to-discrete transfer positive in 88% of architecture-dataset settings. The vulnerability spectrum correlates with power grid contingency rankings (P@10 = 0.66–0.87) as a proof-of-concept cross-domain demonstration. The per-edge ranking directly informs defense design (section V-G).

Limitations. (1) First-order sensitivity thresholds r_v are diagnostic quantities, not probabilistic certificates; they are locally tight for small ε but can be violated at larger perturbation magnitudes (theorem 5). (2) Continuous edge-weight perturbations; discrete insertions/deletions are outside the formal guarantee. Continuous-to-discrete transfer is positive in 29/33 architecture-dataset settings but negative in 4 (12%), particularly for shallow models (GCN-2) and IGNN on dense graphs (table X); practitioners should verify transfer for their specific setting. (3) Standard GAT and architectures using binary adjacency masks require modification for continuous sensitivity; "architecture-agnostic" applies to GNNs with continuous edge-weight-modulated message passing. (4) Power flow $\tau = 0.37$–0.72 is a proof-of-concept; operational deployment requires impedance-aware edge encoding, broader training data, and thermal/voltage violation metrics [43]. (5) The matrix-free pipeline's Neumann series requires $\kappa < 1$; for explicit GNNs without contractivity, the dense path ($N \leq 200$) or finite-difference columns are used. (6) IGNN accuracy (77.5% on Cora) lags behind explicit GNNs (∼82% APPNP); this is the cost of contractivity constraints, not a limitation of S_c. The framework applies identically to higher-accuracy explicit models, which receive the full computational analysis but not the formal regime characterization. (7) 50-node subgraph rankings correlate weakly with full-graph rankings on large citation networks ($\tau = 0.16$ on Cora); the matrix-free full-graph pipeline is recommended for graph-level assessment.

Ethical considerations and dual-use risks. AEGIS identifies maximally sensitive perturbation directions and structurally vulnerable edges, which could be misused to attack deployed GNNs in safety-critical domains—particularly concerning given the power grid case study, where attack optimization tools targeting critical infrastructure demand careful consideration. We argue the diagnostic value outweighs this risk: defenders must understand their system's attack surface to protect it, and the same vulnerability map directly informs defense design (section V-G). Responsible disclosure: we will release code with a tiered access protocol. Vulnerability analysis tools (per-edge rankings, sensitivity thresholds) will be openly available as they are purely defensive. The attack-generation component (SVD-identified perturbation direction) will require institutional affiliation for access, consistent with established practice in adversarial ML [47]. Users of the power grid module should note that vulnerability rankings are conditioned on the GNN model's learned representation, not on physical grid parameters, and should not be used as a substitute for engineering contingency analysis.

Future work. Second-order bounds for formal certificates; discrete perturbation models that extend beyond the continuous threat model; integration with defense mechanisms [13], [12] for vulnerability-aware training with dynamic retraining after edge protection; extending full-graph analysis beyond the current validated boundary (N=7,650 on a single GPU) via distributed JVP computation; edge-feature GNN architectures for power flow applications where line impedance, thermal rating, and voltage level modulate electrical coupling; and investigation of the connection between S_c vulnerability scores and graph curvature-based bottleneck analysis [55].

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